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Transforming a Fragile Protein Helix into an Ultrastable Scaffold via a Hierarchical AI and Chemistry Framework

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eLife Assessment

This **important** work describes systematic computational and experimental approaches to turn a moderately stable α-helical bundle into a very stable fold. The authors advance our understanding of α-helix stabilization providing a convenient framework that has general implications for the protein design field. The main claims have **convincing** support through a sound methodology, with strong specific conclusions for designing mechanically, thermally, and chemically stable α-helical bundles.

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Abstract

The rational design of proteins that maintain structural integrity under concurrent thermal, mechanical, and chemical stress remains a challenge in molecular engineering. We present a hierarchical framework that transforms an α-helical domain into an ultrastable scaffold by integrating AI-guided design with foundational chemical principles. This approach progresses from global architectural reinforcement, using multiple AI tools to create a stabilized four-helix bundle, to local chemical tuning, where AlphaFold3 guides the installation of salt bridges and metal-coordination motifs. A computational pipeline using physics-based screening such as molecular dynamics (MD) simulations efficiently distilled millions of designs into a minimal candidate set. The resulting α-helical proteins exhibit unprecedented multi-axis stability, with mechanical unfolding forces exceeding 200 pN, thermal resilience >100 °C, and high resistance to chemical denaturants. By systematically dissecting the contributions of hydrophobic packing, electrostatics, and metal coordination, we establish a general blueprint for imparting extreme robustness. This work bridges AI-driven structural generation with chemical precision, advancing the creation of durable proteins for mechanistic studies and synthetic biology.

Introduction

Proteins that maintain their structural integrity when heated, stretched, or exposed to harsh solvents are foundational to both life (Manteca, et al., 2017; Neupane, et al., 2024; Rief, et al., 1997) and serve as invaluable building blocks for advanced biomaterials (Fu, et al., 2023; Miserez, et al., 2023; Sivasankaran, et al., 2024), molecular devices (Devaux, et al., 2021; Yang, et al., 2024), and operations in extreme environments (Bustamante, et al., 2004; Pepelnjak, et al., 2024). Among natural scaffolds, helical architectures are particularly appealing targets due to their ubiquity and mechanical functionality: coiled-coils contribute to muscle and other cytoskeletal assemblies (Baumann, et al., 2017; Schwaiger, et al., 2002), talin rod domains act as mechanosensors that unfold under load (del Rio, et al., 2009; Le, et al., 2019;

Tapia-Rojo, et al., 2023), and arrays of α-helical bundles underlie the erythrocyte membrane skeleton (Rief, et al., 1999; Takahashi, et al., 2018; Wensley, et al., 2010). These precedents motivate the goal of transforming common α-helical folds into programmable components with superior, multi-axis stability for diverse applications.

However, the rational design of such multi-faceted stability into isolated helical domains remains challenging (Baker, et al., 2017). The inherent mechanical fragility of the α-helix is a fundamental limitation; under tensile load, its hydrogen bonds rupture sequentially, leading to low unfolding forces of just tens of piconewtons (pN) (Carrion-Vazquez, et al., 2000; Hoffmann, et al., 2013; Law, et al., 2003; LeBlanc, et al., 2021; Muller, et al., 2021; Neuman and Nagy, 2008; Randles, et al., 2007), well below the hundreds of pN reached by mechanically robust β-sheet topologies pulled in shear (Brockwell, et al., 2003). This inherent fragility persists even in the presence of stable hydrophobic cores (Collet, et al., 2005; Craig, et al., 2004; Lee, et al., 2006; Lim, et al., 2008; Milles, et al., 2017; Minin, et al., 2017; Ng, et al., 2007).

While decades of research have identified key chemical determinants of protein stability like hydrophobic packing, electrostatics, and metal coordination, with both natural and engineered examples showing these factors can enhance resilience (Baker, et al., 2015; Goldenzweig and Fleishman, 2018; Kellis, et al., 1988; Marqusee and Baldwin, 1987; Nandi and Ainavarapu, 2022; Wang, et al., 2021; Zhang, et al., 2025), progress in increasing mechanical strength of isolate helical proteins has been a meaningful yet incremental process (Lopez-Garcia, et al., 2021; Tunn, et al., 2018). For example, only a few natural coiled-coils (e.g., in fibrinogen) approach 100 pN, while canonical α-helical bundles such as spectrin and ankyrin repeats typically unfold near 50 pN under comparable loading. The key challenge we address is thus to move beyond adding individual stabilizing features, and instead to precisely combine these different chemical interactions within a single architecture so they work together to resist multiple forms of stress.

The recent rise of generative AI for protein design has dramatically expanded the accessible fold space (Dauparas, et al., 2022; Jumper, et al., 2021; Lin, et al., 2023; Sakuma, et al., 2024; Watson, et al., 2023), offering a path to overcome the architectural limitations of natural scaffolds. However, a critical gap persists. AI models excel at global structure generation but often lack the understanding of local chemical interaction networks required for extreme, multi-axis stability. These models generate vast candidate libraries, creating a screening bottleneck that impedes the experimental validation essential for iterative, chemically rational design. A method that seamlessly integrates generative design with highly efficient screening is needed to bridge this gap.

Here, we treat multi-axis protein stabilization as a problem of hierarchical chemical engineering. We introduce an AI-guided blueprint that progresses systematically from global architectural reinforcement to atomistically precise chemical installation. First, AI-enabled backbone construction generates stable scaffolds with optimized hydrophobic cores, establishing a foundational stability layer. We then apply precision functionalization, installing a minimal set of inter-helical salt bridges and bi-histidine metal-coordination motifs through rational, site-specific edits. This stepwise approach allows us to quantitatively dissect the mechanical contribution of each engineered chemical interaction (Sakuma, et al., 2024).

To make this hierarchical chemical design tractable, we implement a computational pipeline that leverages developability filters (Kyte and Doolittle, 1982), foldability assessment (Jumper, et al., 2021; Lin, et al., 2023), and critically molecular dynamics (MD) simulations (Karplus and McCammon, 2002; Milles, et al., 2018; Rennekamp, et al., 2024) to rank candidates based on proxies for mechanical and thermal resilience. This pipeline compresses $\sim 10^6$ *in silico* designs to a manageable number for experimental validation. The result is a family of ultrastable α-helical proteins where we can quantitatively attribute stability gains to specific chemical engineering steps: hydrophobic core optimization, electrostatic interaction, and metal-coordination clamping. This work provides a general, rational blueprint for the precision engineering of protein energy landscapes, transforming a mechanically weak architecture into a durable platform.

Results

Hierarchical design of ultrastable proteins by integrating AI and chemical principles

To establish a framework for programming multi-axis stability, we developed a hierarchical design strategy that integrates architectural and chemical stabilization principles. We selected the human spectrin repeat R15 (PDB code: 3F57) (Ipsaro, et al., 2009), a three-helix bundle that unfolds at ∼50 pN and melts near 50 °C, as our design template (Figure 1a-b). These moderate baseline properties provide an ideal starting point for quantitatively evaluating the contribution of each engineered stabilization layer. Our approach comprises two distinct stages (Figure 1c): Stage I establishes a stabilized architectural framework through AI-guided backbone construction and computational screening, while Stage II applies precision functionalization with site-specific salt bridges and metal-coordination motifs to reinforce key structural interfaces.

In Stage I, we expanded the native architecture by appending a fourth helix to the spectrin template using RFdiffusion (Figure 1d) (Watson, et al., 2023). We generated 100 backbone variants (50 each for N- and C-terminal extensions) and selected five optimal four-helix scaffolds based on bundle geometry, helical registry, and the potential for hydrophobic core densification without steric clash (Lombardi, et al., 2019). For each selected scaffold, we used ProteinMPNN to generate 100,000 sequences optimized for the remodeled core, producing a total library of 5×10^5 candidate sequences (Liu, et al., 2025).

To identify the best designs fulfilling our stability criteria, we implemented a multi-stage computational funnel that progresses from coarse filtering to high-fidelity functional prediction (Figure 1e). The library was first subjected to a developability screen using physicochemical criteria (GRAVY < −0.3), retaining about 300,000 sequences with high predicted solubility and yield. Foldability was then assessed through a two-tiered model: ESMFold provided an initial screening to 357 designs, which were subsequently refined with AlphaFold2 under strict confidence thresholds (pLDDT > 90, RMSD < 2.0 Å). This process yielded 211 high-confidence designs for further analysis.

Structural characterization revealed that the designed proteins exhibited enhanced packing, with a lower relative solvent-accessible surface area (Porollo, et al., 2013) (rSASA ≈ 0.35) compared to the natural spectrin repeat (rSASA ≈ 0.40). This was further supported by a higher proportion of residues with rSASA < 0.2 (designs: >30% vs. natural spectrin: 23%, see Raw Data 1), indicating successful hydrophobic core optimization (Porollo, et al., 2013).

We then implemented MD simulations to prioritize candidates based on functional stability metrics. Steered MD (SMD) simulations served as a mechanical proxy, stretching each domain under constant-velocity pulling to obtain relative stability rankings (Figure 1b) (Ni, et al., 2024). Complementary annealing MD (AMD) simulations assessed thermal resilience through temperature ramping (310 K to 473 K) while monitoring structural integrity. These physics-based simulations provided the discriminatory power necessary to select candidates exhibiting target multi-axis stability. By integrating SMD and AMD results, we selected a final experimental shortlist that balances predicted mechanical and thermal resilience.

From this refined set, three variants, originally designated SpecAI87788, SpecAI16941, and SpecAI95889 (Figure 1—Figure supplement 1) based on their AI-generated identifiers, were selected for experimental validation based on their superior combined SMD and AMD performance (Figure 1—Figure supplement 2 and Figure 1—Figure supplement 3, Figure 1—video 1-Figure 1—video 10). For clarity in subsequent analyses, these were renamed SpecAI88, SpecAI41, and SpecAI89, reflecting the last two digits of their original IDs.

The funnel compressed $\sim 10^6$ AI-generated sequences to 3 best candidates, preserving both scaffold and sequence diversity while prioritizing top performers. This selection process demonstrates how computational methods can efficiently navigate vast design spaces to identify optimal candidates for experimental characterization.

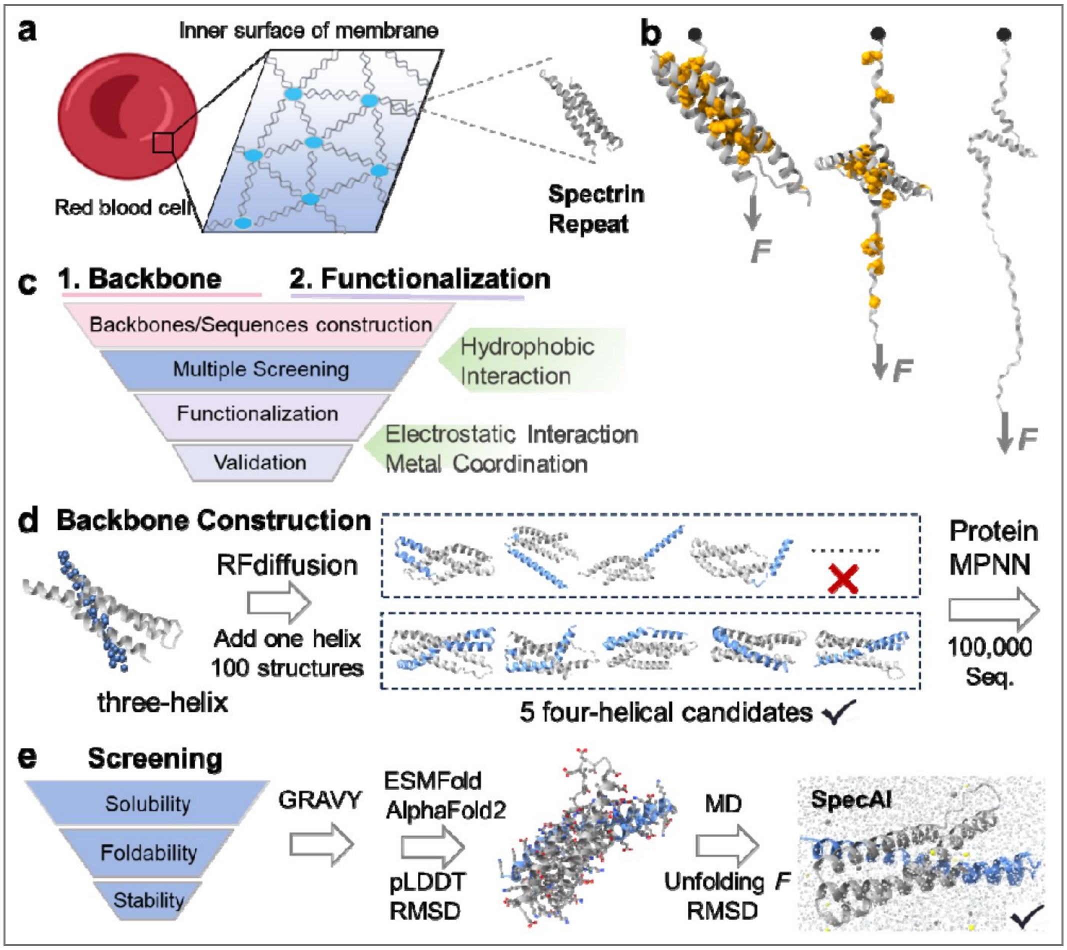

Figure 1. **A hierarchical chemical blueprint for multi-axis stabilization of α-helical protein.**

a, The erythrocyte membrane skeleton, which relies on spectrin repeats to withstand shear stress. Inset: Structure of a single, naturally fragile spectrin repeat (PDB 3F57), with hydrophobic core residues shown. **b**, The mechanical unfolding pathway of a spectrin repeat under tensile force, probed computationally by steered molecular dynamics (SMD) to generate force-extension curves. **c**, Overview of the two-stage design strategy. Stage I (Architectural Stabilization): AI-guided backbone construction and computational screening generate four-helix designs with optimized hydrophobic cores. Stage II (Precision Functionalization): Rational installation of inter-helical salt bridges and metal-coordination motifs reinforces specific mechanical interfaces. **d**, Stage I, Backbone construction. RFdiffusion appends a fourth helix to the native three-helix template, generating 100 initial backbones. Five optimal four-helix scaffolds are selected, and ProteinMPNN is used to generate 100,000 sequences per scaffold. **e**, Stage I, Computational screening. A multi-step funnel prioritizes candidates through successive filters: developability (GRAVY score ≤ −0.3), foldability (ESMFold triage followed by AlphaFold2 refinement with RMSD ≤ 2.0 Å and pLDDT ≥ 90), and stability assessed via molecular dynamics. The process efficiently narrows ~10^6 initial designs to an experimentally tractable shortlist.

Experimental verification of multi-axis stability in designed proteins

From the computational shortlist, we selected three representative designs, SpecAI88, SpecAI41, and SpecAI89, for experimental validation. All three expressed soluble in *E. coli* with high yields (~500 mg/L in TB, Table 1). SDS-PAGE showed the expected band near 19 *k*Da (Figure 2—Figure supplement 1), and mass spectroscopy confirmed molecular weights of 19,760, 19,108 and 19,285 Da, respectively (Figure 2a). Far-UV circular dichroism (CD) spectroscopy was performed, revealing that all three proteins adopted predominantly α-helical structures, as indicated by the characteristic double minima at 208 nm and 222 nm (Figure 2b) (Greenfield, 2007).

To evaluate thermal stability, we performed temperature-dependent circular dichroism (CD) measurements from 20°C to 100°C. All three designed proteins maintained substantial helical structure at 100°C, as evidenced by the retention of characteristic ellipticity at 195 nm, demonstrating exceptional thermodynamic stability (Figure 2b, inset). We further challenged the proteins by performing thermal denaturation in the presence of the chemical denaturant guanidine hydrochloride (GdnHCl) (Liu and Thirumalai, 2025). Notably, all designs preserved their secondary structure at denaturant concentrations up to 3 M and maintained significant helical content under elevated temperature conditions (~50°C, 222 nm), demonstrating remarkable resistance to combined chaotropic stress (Figure 2c). These results confirm not only the proper folding of these monomeric α-helical scaffolds but also validate that our Stage I designs, featuring optimized hydrophobic cores, achieve exceptional stability against both thermal and chemical denaturation.

Next, we directly quantified their mechanical stability using atomic force microscopy-based single-molecule force spectroscopy (AFM–SMFS). Each designed SpecAI domain was fused in series with three GB1 fingerprint domains and pulled via the high-strength Coh–Doc handle (rupture force ~400 pN), enabling unambiguous single-molecule identification (Figure 3a) (Deng, et al., 2019; Shi, et al., 2022; Stahl, et al., 2012). All experiments were conducted in a standard buffer at a pulling speed of 1000 nm/s unless otherwise specified. Force-extension curves were fitted with the worm-like chain (WLC) model to determine the contour length increment (ΔLc) for each unfolding event (Marko and Siggia, 1995).

As expected, the natural spectrin control (102 amino acids in length) exhibited a ΔLc of 32 nm, in addition to the unfolding steps of the three GB1 domains (18 nm each) (Cao, et al., 2006). In contrast, the designed SpecAI proteins showed a significantly larger ΔLc of ≈52 nm (Figure 3b, Figure 1—Figure supplement 1), consistent with the full unraveling of their ~150-residue domain. The measured values: 52 ± 2 nm (SpecAI88), 53 ± 1 nm (SpecAI41), and 52 ± 2 nm (SpecAI89) (mean ± s.e.m., from Gaussian fit), closely matched theoretical predictions, thereby confirming the correct assignment of the designed domain's unfolding (Figure 3c).

Most importantly, all three SpecAI variants exhibited markedly elevated unfolding forces compared to natural spectrin (56 ± 3 pN; n = 212). The unfolding forces reached 116 ± 2 pN (SpecAI88; n = 224), 156 ± 4 pN (SpecAI41; n = 218), and 121 ± 4 pN (SpecAI89; n = 174) (Figure 3c, Table 1). The combination of accurate ΔLc, the presence of three GB1 fingerprint domains, and consistent force statistics confirms single-molecule specificity and establishes a new mechanical baseline for α-helical proteins created through Stage I backbone design.

Precision scaffold functionalization via hierarchical chemical design

Building upon the stabilized architectural frameworks from Stage I, we implemented a precision functionalization strategy to further enhance stability through the rational design of specific chemical interactions at inter-helical interfaces. This stage involved two complementary approaches: engineering of electrostatic salt bridges and incorporation of metal-coordination motifs (Figure 4a).

Protein	SMD (pN) 1 nm/ns	*Tm* (℃)	Yield (mg/L)	AFM (pN)	ΔLc (nm)	Δ*F* (pN)
Spectrin	599±62	~50	~50	56 ± 3	32 ± 1	N/A
SpecAI88	735±80	>100	580 ± 36	116 ± 2	52 ± 2	60
SpecAI41	689±84	>100	625 ± 28	156 ± 4	53 ± 1	100
SpecAI89	670±84	>100	450 ± 16	121 ± 4	52 ± 2	61

Values for SMD are reported as mean ± SD, while all other values are reported as Gaussian fit mean ± s.e.m. Δ*F* means the unfolding force increase over spectrin.

Table 1. **Stability of AI-designed spectrin with four α-helices**

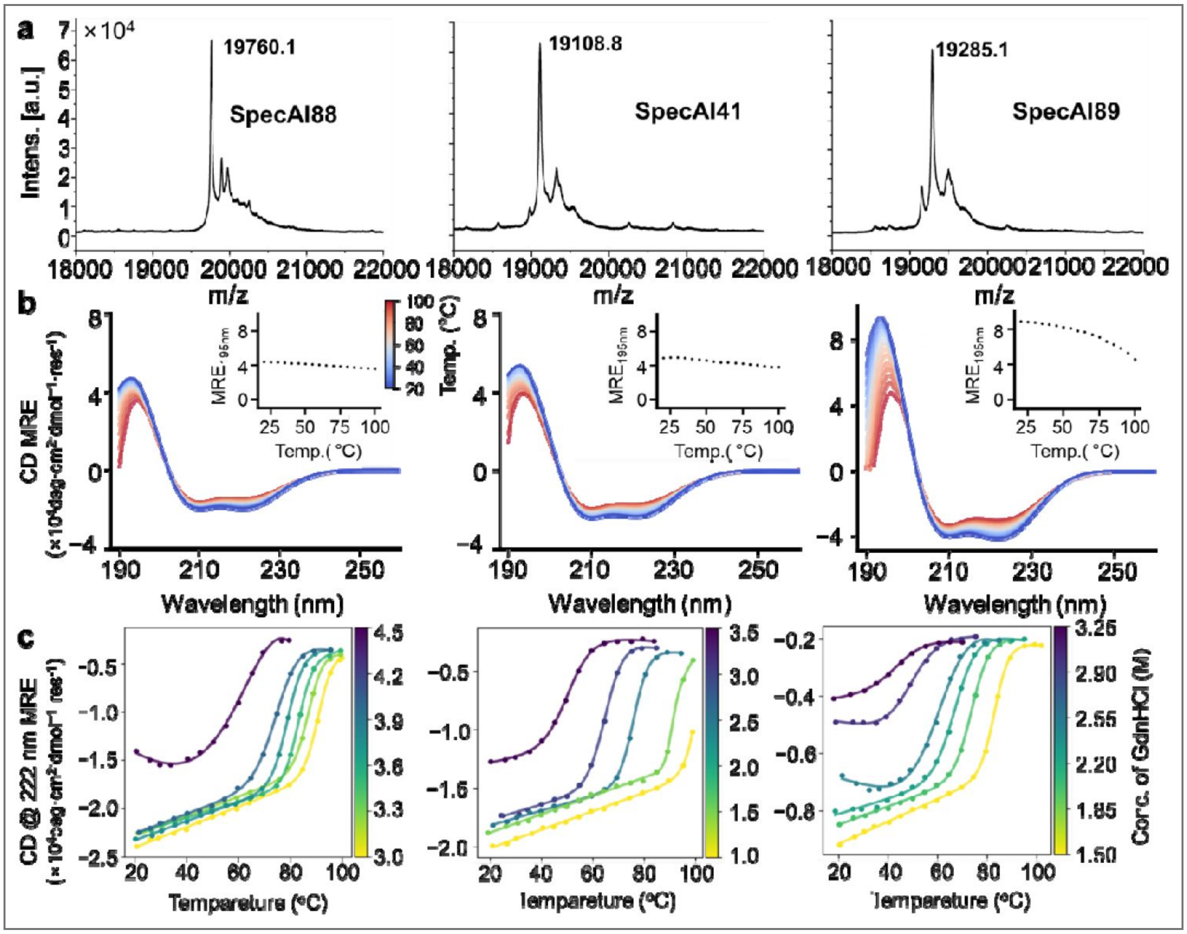

Figure 2. **Thermal and chemical robustness of AI-designed spectrin variants.**

a, MALDI-TOF mass spectrometry confirm molecular weights for SpecAI88, SpecAI41, and SpecAI89. **b**, Far-UV circular dichroism (190–260 nm) shows α-helical signatures with minima at 208 and 222 nm. Temperature-dependent CD (20–100 °C, blue to red) indicates substantial retention of ellipticity at 195 nm, with melting temperatures exceeding 100 °C. **c**, CD at 222 nm recorded in GdnHCl demonstrates persistence of α-helical signal at high denaturant concentrations (~3 M), indicating high chemical resistance.

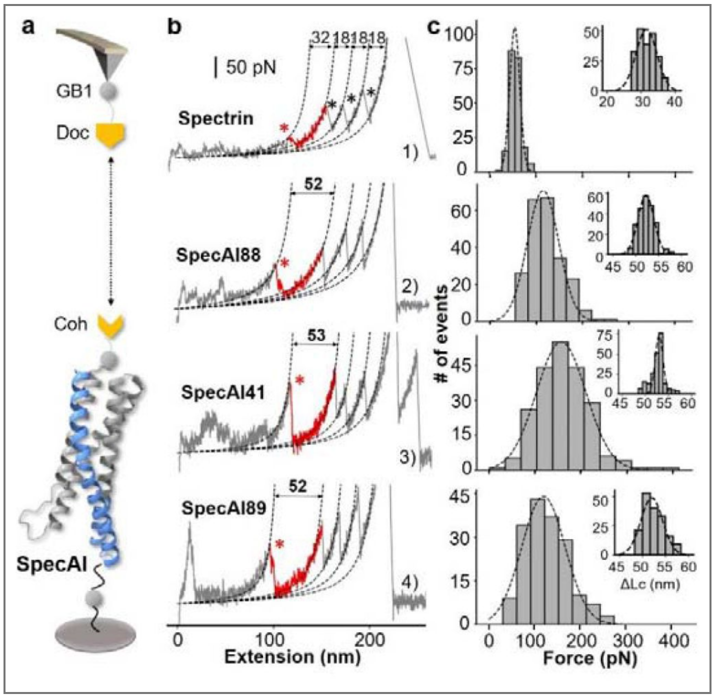

Figure 3. Stage I four-helix bundle designs exhibit enhanced mechanical stability by single-molecule force spectroscopy.

a, Schematic of the AFM–SMFS experimental setup. A dockerin (Doc)-functionalized AFM tip engages a cohesin (Coh)-tagged SpecAI construct immobilized on the surface, enabling single-molecule pulling. The construct includes three GB1 domains as fingerprint markers. **b**, Representative force-extension curves. Curve 1: Unfolding of the natural spectrin repeat (ΔLc ≈ 32 nm, red), followed by three GB1 (ΔLc ≈ 18 nm per domain, black). Curves 2-4: Unfolding of SpecAI variants (ΔLc ≈ 53 nm). Sometimes, the Doc unfolds showing additional peak (Curve 3). Dashed lines are worm-like chain model fit. **c**, Unfolding force histograms of SpecAI (bin size= 30 pN) with Gaussian fits demonstrate a significant increase in mechanical stability compared to the native spectrin repeat (56 ± 3 pN, *n* = 212, bin size=13.75 pN): SpecAI88, 116 ± 2 pN (*n* = 224); SpecAI41, 156 ± 4 pN (*n* = 218) and SpecAI89, 121 ± 4 pN (*n* = 174). The corresponding ΔLc distributions are centered near 52 nm, consistent with the full unfolding of the designed domain.

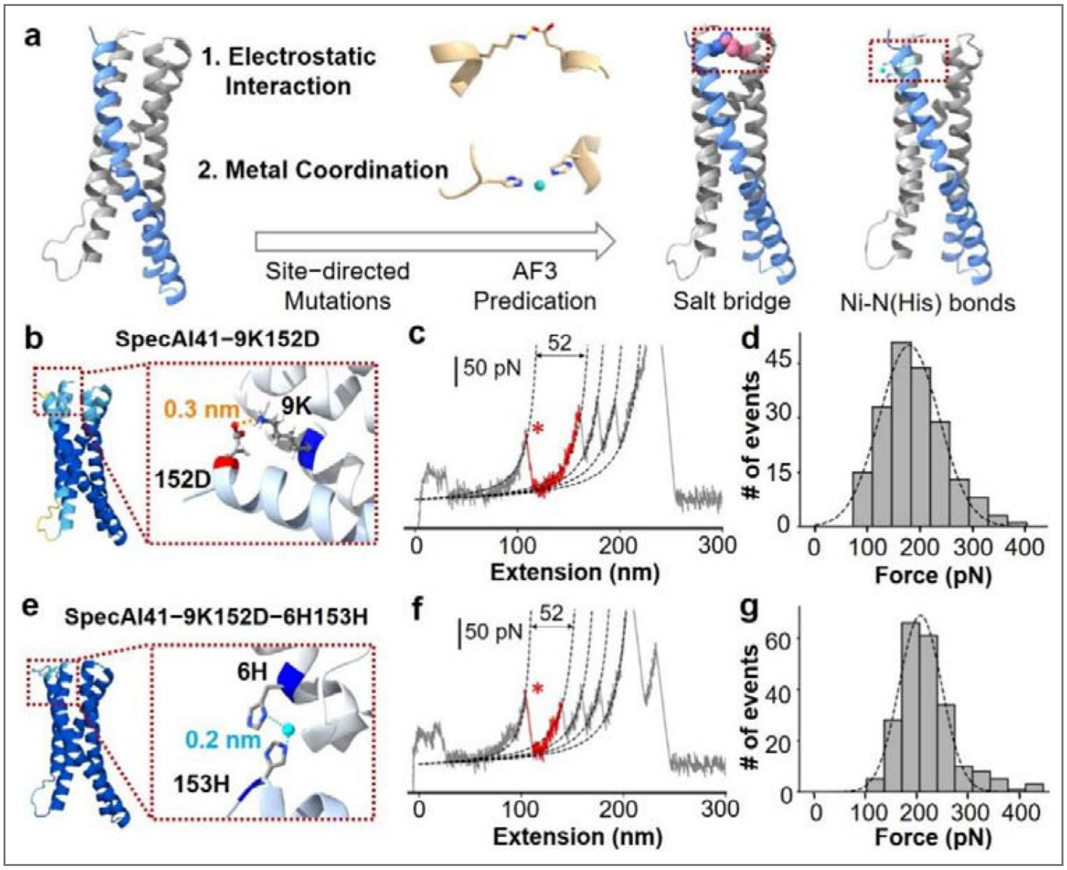

Figure 4. Precision stabilization of designed proteins via electrostatic interactions and metal coordination.

a, Schematic of Stage II design: introducing inter-helical ion pairs and metal-coordination sites into AI-designed backbones to stabilize specific interfaces. **b**, AlphaFold3-predicted structure of variant SpecAI41-9K152D, showing an engineered salt bridge (Lys9-Asp152, 3 Å) designed for electrostatic stabilization without perturbing the core. The color of protein is based on the pLDDT value, showing a high confidence structure prediction (>80). **c**, A representative force-extension curve for the salt-bridge variant shows an unfolding event with a ΔLc of 53 nm, consistent with the parent scaffold. **d**, Unfolding force histograms reveal a ~25 pN increase in mechanical stability for salt-bridge variants compared to their Stage-I parents. **e**, Introduction of a metal-binding site in variant SpecAI41-9K152D-6H153H, with two histidines positioned 2 Å apart, compatible with Ni^{2+} coordination. **f**, A representative force-extension curve recorded in 200 μM Ni^{2+} shows the unfolding event (ΔLc ≈ 53 nm). **g**, Unfolding force histograms confirm enhanced mechanical stability, with forces reaching ~200 pN, a significant gain over the salt-bridge parent.

Our design strategy followed a rigorous hierarchical selection process. First, we identified potential mutation sites by excluding residues critical for hydrophobic core integrity, focusing instead on positions at inter-helical interfaces that could accommodate new interactions without perturbing the existing structural stability. Second, we employed AlphaFold3 (AF3) structure prediction to evaluate potential mutations, requiring that designs not only maintain high overall structural confidence (pLDDT > 80 for all mutated residues, with many exceeding 90 but also satisfy precise geometric criteria for the intended molecular interactions, specifically in terms of inter-atomic distances.

We designed and incorporated single inter-helical salt bridges into each optimized parent scaffold: SpecAI41-9K152D (Figure 4b, Figure 1—Figure supplement 1),

SpecAI88-49E102K, and SpecAI89-25E48K (Figure 4—Figure supplement 3). AF3 models confirmed that the engineered residues were positioned at interface-facing turns with favorable i↔i' or i±1 helical register arrangements without introducing steric clashes. Crucially, all designed salt bridges showed charged atom distances <0.4 nm in the predicted structures, and the contacts remained stable following backbone relaxation simulations. This consistency indicates genuine local interface reinforcement rather than global structural reorganization. All mutated backbone atoms maintained high pLDDT scores (>80), confirming the preservation of structural integrity. For example, the pLDDT scores for the Cα atoms were 90.1 and 80.7 for SpecAI41-9K152D.

AFM-SMFS measurements revealed ~30 pN increases relative to Stage-I parents with ΔLc unchanged (~53 nm), as expected. Unfolding forces were 180 ± 4 pN (n = 197) for SpecAI41-9K152D (Figure 4c–d), 141 ± 3 pN (n = 202) for SpecAI88-49E102K, and 153 ± 4 pN (n = 264) for SpecAI89-25E48K (Figure 4—Figure supplement 3, Table 2). The constant ΔLc across parents and salt-bridge variants indicates that electrostatic installation does not change topology or domain length, but raises the mechanical barrier along the established pathway.

To achieve additional mechanical reinforcement, we designed bi-histidine motifs capable of forming intra-domain Ni(II)-coordination sites[63]. These motifs were strategically positioned on adjacent helices to create bidentate metal-binding pockets without perturbing the hydrophobic core. Our design criteria targeted Ni-N(His) distances of ~2.0 Å, Cα-Cα spacing compatible with inter-helical bridging rotamers, and proper side-chain orientation (Nδ1/Nε2) for optimal imidazole coordination geometry. This approach yielded three metal-coordination variants: SpecAI41-9K152D-6H153H, SpecAI88-49E102K-6H149H and SpecAI89-25E48K-24H135H (Figure 4, Figure 1—Figure supplement 1 and Figure 4—Figure supplement 3), whose structures were validated using AF3 again.

In the presence of 200 μM Ni^{2+}, AFM-SMFS revealed a further significant increase in mechanical stability over the salt-bridge parents, with unfolding forces reaching 208 ± 3 pN (n = 222) for SpecAI41-9K152D-6H153H, 190 ± 3 pN (n = 232) for SpecAI88-49E102K-6H149H, and 173 ± 3 pN (n = 204) for SpecAI89-25E48K-24H135H, and an unchanged ΔLc of ~53 nm (Table 2). These results are consistent with a short, inner-interface coordination clamp that reinforces the native unfolding pathway without introducing alternative topologies.

We further characterized the unfolding kinetics of the computationally designed proteins via dynamic force spectroscopy across a range of pulling speeds, focusing on the SpecAI41 series due to its highest unfolding force. The unfolding forces of the three SpecAI41 exhibited the expected linear dependence on the logarithm of the loading rate (Figure 4—Figure supplement 4). Fitting the data to the Bell-Evans model yielded a consistent distance to the transition state ($\Delta x \approx 0.19$ nm) for all three constructs, while the spontaneous unfolding rate (k_{off}) decreased progressively from SpecAI41 (0.44 s^{-1}) to the salt-bridge (0.09 s^{-1}) and metal-coordination (0.05 s^{-1}) variants. This kinetic profile further confirms that the engineered stabilizations raise the activation barrier for unfolding without altering the unfolding pathway.

Similarly, we assessed the thermal stability of the Stage II variants (Figure 4—Figure supplement 5). The results demonstrate that the engineered salt bridges and metal-binding sites maintained excellent thermodynamic stability with T_m above 100 °C. In chemical stability tests, the Stage II

Protein	Force (pN)	ΔLc (nm)	ΔF (pN)	Total ΔF
SpecAI88-49E102K	141 ± 3	54 ± 1	25	74
+6H149H	190 ± 3	53 ± 1	49	
SpecAI41-9K152D	180 ±□4	53 ±□2	24	52
+6H153H	208□±□3	53 ±□2	28	
SpecAI89-25E48K	153 ±□4	53 ± 1	32	52
+24H135H	173 ± 3	53 ± 1	20	

Total ΔF means the unfolding force increase over SpecAI.

Table 2. **SpecAI stability further enhanced by site-specific functionalization**

variants still retained considerable helical content under elevated temperature but their tolerance to GdnHCl decreased slightly compared to the Stage I variants.

This hierarchical computational design methodology (Figure 5), progressing from architectural stabilization to precision functionalization, enabled the predictable enhancement of protein stability through the additive contribution of distinct chemical interaction networks.

Discussion

The rational design of proteins capable of withstanding concurrent thermal, chemical, and mechanical stress remains a fundamental challenge. Here, we present a hierarchical strategy that integrates chemically distinct stabilization mechanisms within a single protein architecture. Our approach not only provides a robust blueprint for creating ultrastable proteins but also decouples the quantitative contributions of hydrophobic packing, electrostatic interactions, and metal coordination to mechanical resilience.

Our work builds upon established principles of α-helical protein stability while introducing substantive advances. Whereas most isolated α-helical domains unfold below 20 pN due to sequential hydrogen bond rupture (Wolny, et al., 2014), our designed variants achieve forces up to 200 pN. This represents an advance beyond canonical α-helical bundles such as spectrin and ankyrin repeats, which typically unfold near 50 pN (Serquera, et al., 2010). Previous engineering strategies have demonstrated promising improvements potential (Lopez-Garcia, et al., 2021), and our contribution lies in the systematic integration of global architectural optimization with local chemical reinforcement within a unified design framework enabled by AI protein design and computational screening (Thomas and Elcock, 2004; Wolny, et al., 2017). The hierarchical organization of our approach proved effective. We first established a stable structural framework through computational design of an optimized hydrophobic core, providing baseline mechanical stability of ~120 pN. Subsequent precision functionalization with inter-helical salt bridges contributed an additional ~30 pN, while metal coordination sites provided further gains of ~50 pN depending on geometric parameters. This modular strategy demonstrates that distinct chemical interaction networks can be engineered to operate additively within a single protein structure (Cao, et al., 2011).

Notably, conventional structural metrics such as pLDDT scores, while predictive of foldability, show limited correlation with mechanical stability. This divergence underscores that dynamic functional properties require physics-based simulation. Although steered molecular dynamics (SMD) overestimates absolute unfolding forces due to computationally mandated high pulling speeds (Merkel, et al., 1999) (e.g., ~600 pN *in silico* versus ~50 pN for spectrin in experiment), it served as a powerful semi-quantitative screen. Together, these results indicate that SMD simulations provide a useful qualitative reference for comparing relative mechanostability among designs and for guiding experimental prioritization, despite intrinsic differences in force scales and sampling between simulations and experiments.

In conclusion, our hierarchical framework provides a generalizable route to programming protein stability. By modular integration of architectural and chemical stabilization strategies, we can now tailor mechanical, thermal, and chemical resistance in α-helical proteins. This approach will accelerate the development of robust protein-based materials and molecular devices for demanding applications in biotechnology and synthetic biology (Klein and Nanda, 2025; Ling, et al., 2018).

Materials and methods

Plasmids construction and proteins engineering

The gene fragments encoding the designed proteins were synthesized (General Biosystems, Anhui) and cloned into the pET-30a vector in the format *Coh-GB1-SpecAI-GB1-His$_6$-NAL* for use in AFM-SMFS experiments (Shi, et al., 2022). All point mutation constructs introducing electrostatic and

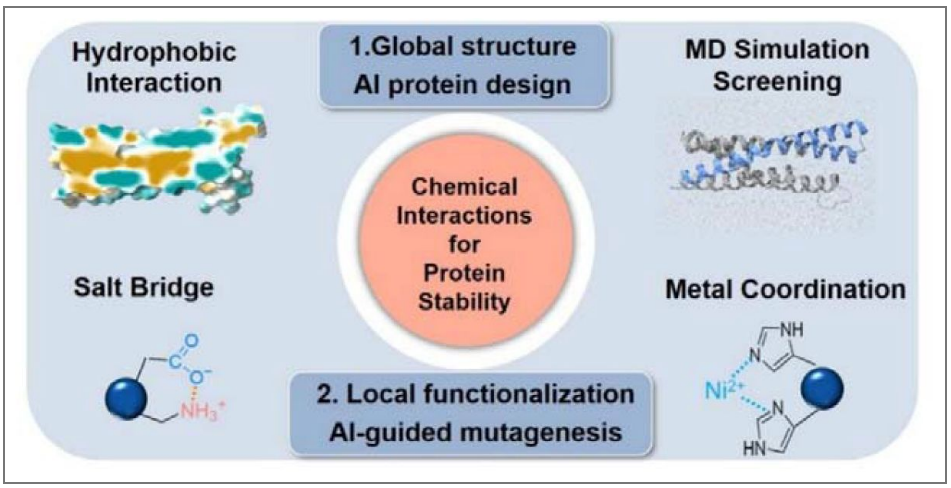

Figure 5. **Hierarchical computational design of ultrastable proteins through multi-scale stabilization.**

Schematic summarizing the two-stage design strategy for additive mechanical reinforcement. Stage I establishes a stable architectural framework through MD screening of AI-designed hydrophobically optimized cores. Stage II introduces precision functionalization via inter-helical salt bridges and bi-histidine metal-coordination motifs, guided by AlphaFold3 structural models. The integration of global architectural stability with local chemical cross-linking produces additive mechanical reinforcement, enabling the creation of ultrastable protein domains.

metal-ligand interactions were generated by PCR-based site-directed mutagenesis. A construct, *pET30a-SpecAI-His*$_6$, was prepared via seamless cloning for circular dichroism (CD) measurements.

Both protein constructs were expressed and purified following identical protocols. The plasmids were transformed into *Escherichia coli* BL21(DE3) and plated on Luria-Bertani (LB) agar medium (2.5% LB, 2% agar) containing 100 µg/mL kanamycin. Single colonies were inoculated into 20 mL of LB broth supplemented with kanamycin and incubated overnight at 37°C. These were subsequently expanded into 800 mL of LB medium and grown until the optical density at 600 nm (OD_{600}) reached 0.7–0.8. After cooling to room temperature, protein expression was induced by adding 0.5 mM IPTG, followed by incubation at 18°C for 20 hours. Cells were harvested by centrifugation at 4,000 rpm for 10 minutes and stored at −80°C prior to purification.

For purification, cell pellets were resuspended in 30 mL of lysis buffer (50 mM Tris, pH 7.4) supplemented with DNase I, RNase A, and PMSF. Bacterial disruption was carried out using a high-pressure homogenizer (700 bar, 3 minutes), and the lysate was centrifuged at 13,000 rpm for 15 minutes to collect the supernatant. This was incubated with Co-NTA affinity resin pre-equilibrated with binding buffer (20 mM Tris, 400 mM NaCl, 2 mM imidazole, pH 7.4) for 30 minutes at room temperature. After washing with 3–5 column volumes of binding buffer, the target protein was eluted with 15 mL of elution buffer (20 mM Tris, 400 mM NaCl, 250 mM imidazole, pH 7.4).

The *Coh-GB1-SpecAI-GB1-His*$_6$*-NAL* protein was concentrated via ultrafiltration and buffer-exchanged into a low-salt storage buffer (50 mM Tris, 100 mM NaCl, pH 7.4). The *SpecAI-His*$_6$ variant was similarly processed and exchanged into a CD assay buffer (10 mM K_2HPO_4, 1.8 mM KH_2PO_4, pH 7.4).

The *GL-His*$_6$*-GB1-Xmod-Doc*, *GL-His*$_6$*-ELP20-Cys* and *His*$_6$*-Cys-ELP20-NGL* protein used for subsequent single-molecule force spectroscopy experiments was expressed and purified using a similar procedure. The expression and purification protocols of *Oa*AEP1(C247A) can be found in references (Deng, et al., 2019; Yang, et al., 2017).

Structural modeling and MD simulations

First, based on spectrin structure, a total of 100 different protein structures were generated using RFdiffusion by appending new structural elements to either the N- or C-terminus (Watson, et al., 2023). Among these, five structurally reasonable models exhibiting four helical topologies were manually selected for their potential to improve hydrophobic interactions. For each selected structure, 100,000 sequence variants were generated using ProteinMPNN, using a fixed random seed of 37 to ensure reproducibility (Dauparas, et al., 2022). 319,747 sequences with GRAVY score below –0.3 were selected. The structural compatibility of the generated sequences was then quickly assessed using ESMFold and compared with the corresponding initial structures predicted by Rfdiffusion (Lin, et al., 2023). Only 357 Sequences exhibiting high structural agreement, defined by pLDDT scores above 90 and root mean square deviation (RMSD) values below 2.0 Å, were further modeled using AlphaFold2 (Jumper, et al., 2021). During AlphaFold2 modeling, six recycles were performed, with template mode enabled. For each sequence, five structural models were generated and ranked according to their pLDDT scores. Ultimately, only the ones with significantly higher scores were retained, and models with pLDDT >90 and RMSD < 2 Å were selected for subsequent analysis, resulting in a total of 211 models.

To assess the mechanical stability of designed proteins, Steered MD (SMD) simulations were conducted by GROMACS 2023.2 (single-precision build) (Van Der Spoel, et al., 2005). The systems were constructed using GROMACS built-in tools, with the CHARMM36m force field and the TIP3P water model applied (Lee, et al., 2015). Rectangular simulation boxes were used with a minimum protein-box distance of 1.2 nm in all directions; for SMD simulations, the *z* dimension was extended to the theoretical stretching length (initial N-C distance plus 0.36 nm per stretched residue). All simulations were performed assuming a physiological pH of 7.0, with standard protonation states assigned accordingly. The systems were neutralized with sodium and chloride ions to achieve a physiological NaCl concentration of 0.15 M. Simulations were equilibrated under

NVT conditions at 310 K using the V-rescale thermostat, followed by a 5 ns NpT production run with C-rescale pressure coupling (1.0 bar), while maintaining temperature control as described above (Kim, et al., 2019).

Prior to SMD simulations, each system was first subjected to 5,000 steps of energy minimization using the steepest descent algorithm, with all protein backbone atoms positionally restrained. Subsequently, during a 1.0 ns NVT equilibration, these restraints were maintained with a harmonic force constant of 1000 $kJ \cdot mol^{-1} \cdot nm^2$ to allow for the thorough relaxation of the solvent and ions around the protein. After this, 125 ps equilibrium MD runs and 5 ns production MD simulations were run to stabilize the temperature and pressure, ensuring system equilibrium. During SMD simulations, the N-terminus of each protein was restrained using a harmonic potential with a force constant of 1000 $kJ \cdot mol^{-1} \cdot nm^2$, while the C-terminus was pulled along the z-axis at constant velocity using an umbrella restraint with the same force constant (k = 1000 $kJ \cdot mol^{-1} \cdot nm^2$).

SMD simulations were performed at a pulling speed of 1 nm/ns on the 211 predicted protein structures selected from the previous design step (Gräter, et al., 2005; Lu, et al., 1998). Each simulation was repeated three times to ensure reproducibility. Since the designed helical proteins could unfold at multiple sites along their structure, the total extension distance was limited to 65% of the theoretical full length, a value that was previously validated through full-length stretching simulations of individual proteins, which confirmed that no significant unfolding events occurred beyond this distance. During SMD simulations, the C-terminus was harmonically restrained while a harmonic pulling potential was applied to the N-terminus in umbrella mode, directed along the z-axis. The pulled group was unconstrained in the xy-plane, allowing for lateral relaxation during extension. The top three proteins exhibiting the highest average unfolding forces in SMD simulations were selected as candidate constructs. To validate their thermal stability, annealing MD (AMD) simulations were conducted on each protein, with temperature gradually increased from 310 to 473 K over a 1 μs trajectory using a 2 fs time step. Structural deviation was quantified by calculating the RMSD between the final structure and the original predicted conformation. These AMD simulations served to validate the selected proteins, ensuring that the mechanically robust candidates also exhibit favorable thermal stability for downstream applications.

AFM-SMFS experiments

AFM-SMFS protein unfolding experiments were performed on a ForceRobot300 AFM (JPK) at room temperature. Detailed protocols are provided in the literature (Liu, et al., 2024). In brief, proteins were immobilized on both glass substrate and AFM cantilever via strain-promoted azide-alkyne cycloaddition (SPAAC) and Michael addition chemistry (Deng, et al., 2019; Shi, et al., 2022). Elastin-like polypeptide (ELP20, $(VPGXG)_{20}$) was employed as a spacer and single-molecule signature for the experiment (Ott, et al., 2017). Prior to the measurements, ELP20 with an *Oa*AEP1(C247A) specific recognition site C-terminal NGL (or NAL) or N-terminal GL (*His*$_6$*-Cys-ELP20-NGL/NAL*, *GL-His*$_6$*-ELP20-Cys*) were separately immobilized on the AFM tips and glass substrates through maleimide-thiol reactions. *Oa*AEP1 catalyzes the formation of a covalent bond between a C-terminal -NGL/-NAL tag and an N-terminal -GL motif (Deng, et al., 2019; Yang, et al., 2017). Then GB1-Doc (GB1-Xmodule-dockerin) was linked at cantilevers by enzymatic ligation using *Oa*AEP1 to provide X-module-dockerin (Doc) domains (Stahl, et al., 2012). *Coh-GB1-SpecAI-GB1-His*$_6$*-NAL* fusion protein was immobilized on a modified glass. The AFM tip was pressed onto the surface with a force of 400 pN to capture the target protein via the specific and high-affinity Coh–Doc interaction, enabling reliable single-molecule attachment.

AFM-SMFS measurements were conducted in buffer composed of 50 mM Tris-HCl and 100 mM NaCl at pH 7.4. For proteins engineered with metal coordination sites, 200 μM $NiCl_2$ was supplemented to facilitate Ni^{2+} binding. The tip was retracted at a constant pulling speed of 1000 nm/s if not specified. For dynamic force spectroscopy experiments, pulling speeds of 400, 1000, 2000, and 4000 nm/s were used.

Bruker MLCT-D or MLCT-E cantilevers were used, and each cantilever was calibrated in buffer prior to data acquisition using the thermal noise method (equipartition theorem) via the instrument's built-in software. These curves were analyzed by Igor Pro 6.12 (Wavemetrics). Only traces exhibiting at least three consecutive GB1 unfolding peaks followed by a final high force Coh–Doc rupture peak were selected for further analysis. Unfolding forces and contour length increments (ΔLc) were extracted by fitting each individual peak using the worm-like chain (WLC) model of polymer elasticity. Additionaly, the measured unfolding forces were further calibrated using the fingerprint protein GB1 domain as an internal standard, based on its well-established unfolding force (213 pN at 1000 nm/s) (Cao, et al., 2006). For each construct, unfolding forces from multiple independent traces were pooled and fitted with a Gaussian function. The mean unfolding force and the standard error of the mean (SEM) were extracted from the fits and used for subsequent comparison and analysis.

To investigate the effect of loading rate on protein unfolding behavior, AFM measurements were performed not only at the retraction speed of 1000 $nm{\cdot}s^{-1}$ but also at constant cantilever retraction speeds of 400, 2000, and 4000 $nm{\cdot}s^{-1}$. For each pulling speed, multiple force–extension curves were collected. For each unfolding event, the loading rate r was calculated as the product of the slope of the force peak and the cantilever retraction speed, and calibrated using the unfolding force of the GB1 domain as an internal standard (Cao, et al., 2006). For each speed, the average r and peak unfolding force F were determined, with standard deviations reported as error bars, and plotted in a force–loading rate scatter plot. The zero-force off-rate (k_{off}) and the distance to the transition state (Δx) were subsequently extracted by linear fitting based on the Bell–Evans model (Evans and Ritchie, 1997):

$$F = \frac{k_B T}{\Delta x} \ln\left(\frac{\Delta x}{k_{\text{off}} k_B T}\right) + \frac{k_B T}{\Delta x} \ln(r)$$

where k_B is the Boltzmann constant and T is the temperature.

Site-directed mutagenesis

To further improve the mechanical stability of helical proteins, salt bridge mutations were introduced into the top three candidates from Stage I. Two mutation strategies were employed: introducing salt bridges either at the N- or C-terminus of the protein, or near the regions where force peaks were observed in the SMD simulation. Both strategies targeted areas subjected to mechanical stress. Subsequently, we explored metal coordination as an additional stabilization strategy. Double-histidine mutations were introduced at the N- and C-termini of the parent proteins that exhibited high unfolding forces in previous experiments. Structural predictions using AlphaFold3 confirmed that the overall conformations remained stable and that the histidine residues were properly positioned to satisfy the geometric requirements for Ni^{2+} coordination. These mutated proteins were then subjected to single-molecule force spectroscopy experiments following the same protocol described above.

MALDI-TOF MS

Measurements were performed on an ultrafleXtreme mass spectrometer (Bruker Daltonics). Protein samples were prepared using a two-layer matrix method at a concentration of approximately 3 mg/mL. Each sample was analysed in at least three independent replicates to ensure data reliability. Mass spectra were acquired in the m/z range of 18,000–22,000.

Circular dichroism

CD measurements were performed on a Chirascan (Applied Photophysics) instrument using a 1 mm pathlength cuvette (Hellma). The purified protein was diluted to 0.1–0.15 mg/mL in CD assay buffer. Far-UV CD spectra were recorded from 190 nm to 260 nm at 20°C to evaluate the secondary structure of SpecAI proteins. To assess thermal stability, temperature-dependent CD measurements were conducted, with spectra collected at 5°C intervals from 20°C to 100°C. Thermal unfolding was monitored by tracking the mean residue ellipticity (MRE) at 195 nm (a characteristic positive peak for α-helices) across a temperature gradient from 20°C to 100°C.

For the chemical denaturation study of SpecAI, thermal scanning was conducted in the presence of GdnHCl at concentrations ranging from 1 M to 5 M. A set of 4–6 concentrations was selected based on their ability to reliably determine the sample's *Tm* and produce high-quality fitting curves. Thermal denaturation was performed by heating from 20°C upward in 5°C increments until complete denaturation was achieved. The terminal temperature varied depending on both the specific sample and the concentration of GdnHCl.

At each temperature plateau, full spectra were recorded between 200 nm and 260 nm. Changes in signal were monitored at approximately 222 nm across all temperatures and GdnHCl concentrations. All measurements were repeated in at least three independent replicates to ensure reproducibility.

Data availability

All data are present in the main text or supplementary information.

Acknowledgements

The numerical calculations in this work have been done on the computing facilities in the High-Performance Computing Center (HPCC) of Nanjing University.

Additional files

Supplementary information

Supplementary data

Supplementary video 1

Supplementary video 2

Supplementary video 3

Supplementary video 4

Supplementary video 5

Supplementary video 6

Supplementary video 7

Supplementary video 8

Supplementary video 9

Supplementary video 10

Additional information

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References

Baker EG, Bartlett GJ, Crump MP, Sessions RB, Linden N, Faul CFJ, Woolfson DN (2015) Local and macroscopic electrostatic interactions in single α-helices. *Nat. Chem. Biol* **11**:221-228

Baker EG, Williams C, Hudson KL, Bartlett GJ, Heal JW, Porter Goff KL, Sessions RB, Crump MP, Woolfson DN (2017) Engineering protein stability with atomic precision in a monomeric miniprotein. *Nat. Chem. Biol* **13**:764-770

Baumann F, Bauer MS, Rees M, Alexandrovich A, Gautel M, Pippig DA, Gaub HE (2017) Increasing evidence of mechanical force as a functional regulator in smooth muscle myosin light chain kinase. *eLife* **6**:e26473 https://doi.org/10.7554/eLife.26473

Brockwell DJ, Paci E, Zinober RC, Beddard GS, Olmsted PD, Smith DA, Perham RN, Radford SE (2003) Pulling geometry defines the mechanical resistance of a beta-sheet protein. *Nat. Struct. Mol. Biol* **10**:731-737

Bustamante C, Chemla YR, Forde NR, Izhaky D. (2004) Mechanical processes in biochemistry. *Annu. Rev. Biochem* **73**:705-748

Cao Y, Lam C, Wang M, Li H. (2006) Nonmechanical protein can have significant mechanical stability. *Angew Chem Int* **45**:642-645

Cao Y, Li YD, Li H. (2011) Enhancing the mechanical stability of proteins through a cocktail approach. *Biophysical Journal* **100**:1794-1799

Carrion-Vazquez M, Oberhauser AF, Fisher TE, Marszalek PE, Li HB, Fernandez JM (2000) Mechanical design of proteins-studied by single-molecule force spectroscopy and protein engineering. *Prog. Biophys. Mol. Biol* **74**:63-91

Collet J-P, Shuman H, Ledger RE, Lee S, Weisel JW (2005) The elasticity of an individual fibrin fiber in a clot. *Proc. Natl. Acad. Sci. U. S. A* **102**:9133-9137

Craig D, Gao M, Schulten K, Vogel V. (2004) Tuning the Mechanical Stability of Fibronectin Type III Modules through Sequence Variations. *Structure* **12**:21-30

Dauparas J, Anishchenko I, Bennett N, Bai H, Ragotte RJ, Milles LF, Wicky BIM, Courbet A, de Haas RJ, Bethel N, *et al.* (2022) Robust deep learning–based protein sequence design using ProteinMPNN. *Science* **378**:49-56

del Rio A, Perez-Jimenez R, Liu R, Roca-Cusachs P, Fernandez JM, Sheetz MP (2009) Stretching single talin rod molecules activates vinculin binding. *Science* **323**:638-641

Deng Y, Wu T, Wang M, Shi S, Yuan G, Li X, Chong H, Wu B, Zheng P. (2019) Enzymatic biosynthesis and immobilization of polyprotein verified at the single-molecule level. *Nat. Commun* **10**:2775-2785

Devaux F, Li X, Sluysmans D, Maurizot V, Bakalis E, Zerbetto F, Huc I, Duwez A-S. (2021) Single-molecule mechanics of synthetic aromatic amide helices: Ultrafast and robust non-dissipative winding. *Chem* **7**:1333-1346

Evans E, Ritchie K. (1997) Dynamic strength of molecular adhesion bonds. *Biophys J* **72**:1541-1555

Fu L, Li L, Bian Q, Xue B, Jin J, Li J, Cao Y, Jiang Q, Li H. (2023) Cartilage-like protein hydrogels engineered via entanglement. *Nature* **618**:740-747

Goldenzweig A, Fleishman SJ (2018) Principles of Protein Stability and Their Application in Computational Design. *Annu. Rev. Biochem* **87**:105-129

Gräter F, Shen J, Jiang H, Gautel M, Grubmüller H. (2005) Mechanically Induced Titin Kinase Activation Studied by Force-Probe Molecular Dynamics Simulations. *Biophys. J* **88**:790-804

Greenfield NJ (2007) Using circular dichroism spectra to estimate protein secondary structure. *Nat. Protoc* **1**:2876-2890

Hoffmann T, Tych KM, Hughes ML, Brockwell DJ, Dougan L. (2013) Towards design principles for determining the mechanical stability of proteins. *Phys. Chem. Chem. Phys* **15**:15767-15780

Ipsaro JJ, Huang L, Mondragón A. (2009) Structures of the spectrin-ankyrin interaction binding domains. *Blood* **113**:5385-5393

Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, Tunyasuvunakool K, Bates R, Žídek A, Potapenko A, *et al.* (2021) Highly accurate protein structure prediction with AlphaFold. *Nature* **596**:583-589

Karplus M, McCammon JA (2002) Molecular dynamics simulations of biomolecules. *Nat. Struct. Biol* **9**:646-652

Kellis JT, Nyberg K, Da Săil, Fersht AR (1988) Contribution of hydrophobic interactions to protein stability. *Nature* **333**:784-786

Kim M, Kim E, Lee S, Kim JS, Lee S. (2019) New Method for Constant-NPT Molecular Dynamics. *J. Phys. Chem. A* **123**:1689-1699

Klein D, Nanda V. (2025) The proteins that could be. *Nat. Rev. Chem* **9**:283-284

Kyte J, Doolittle RF (1982) A simple method for displaying the hydropathic character of a protein. *J. Mol. Bio* **157**:105-132

Law R, Carl P, Harper S, Dalhaimer P, Speicher DW, Discher DE (2003) Cooperativity in Forced Unfolding of Tandem Spectrin Repeats. *Biophys. J* **84**:533-544

Le S Yu M, Yan J. (2019) Direct single-molecule quantification reveals unexpectedly high mechanical stability of vinculin-talin/alpha-catenin linkages. *Sci. Adv* **5**:eaav2720

LeBlanc MA, Fink MR, Perkins TT, Sousa MC (2021) Type III secretion system effector proteins are mechanically labile. *Proc. Natl. Acad. Sci. U. S. A* **118**:e2019566118

Lee G, Abdi K, Jiang Y, Michaely P, Bennett V, Marszalek PE (2006) Nanospring behaviour of ankyrin repeats. *Nature* **440**:246-249

Lee J, Cheng X, Swails JM, Yeom MS, Eastman PK, Lemkul JA, Wei S, Buckner J, Jeong JC, Qi Y, *et al.* (2015) CHARMM-GUI Input Generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM Simulations Using the CHARMM36 Additive Force Field. *J. Chem. Theory Comput* **12**:405-413

Lim BBC, Lee EH, Sotomayor M, Schulten K. (2008) Molecular Basis of Fibrin Clot Elasticity. *Structure* **16**:449-459

Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, Smetanin N, Verkuil R, Kabeli O, Shmueli Y, *et al.* (2023) Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**:1123-1130

Ling S, Kaplan DL, Buehler MJ (2018) Nanofibrils in nature and materials engineering. *Nat. Rev. Mater* **3**:18016

Liu L, Zheng B, Ao M, Li J, Wang Z, Jiang Y, Zheng P. (2025) De Novo Design Mechanostable Titin Ig Domain Using Artificial Intelligence and Steered Molecular Dynamics Simulations. *CCS Chem* **7**:1629-1640

Liu Y, Song D, Li S, Guo Z, Zheng P. (2024) Click Chemistry-Based Force Spectroscopy Revealed Enhanced Binding Dynamics of Phosphorylated HMGB1 to Cisplatin-DNA. *J. Am. Chem. Soc* **146**:13126-13132

Liu Z, Thirumalai D. (2025) Impact of Guanidinium Hydrochloride on the Shapes of Prothymosin-α and α-Synuclein Is Dramatically Different. *Biochemistry* **64**:105-113

Lombardi A, Pirro F, Maglio O, Chino M, DeGrado WF (2019) De Novo Design of Four-Helix Bundle Metalloproteins: One Scaffold, Diverse Reactivities. *Acc Chem Res* **52**:1148-1159

Lopez-Garcia P, de Araujo AD, Bergues-Pupo AE, Tunn I, Fairlie DP, Blank KG (2021) Fortified Coiled Coils: Enhancing Mechanical Stability with Lactam or Metal Staples. *Angew Chem Int* **60**:232-236

Lu H, Isralewitz B, Krammer A, Vogel V, Schulten K. (1998) Unfolding of Titin Immunoglobulin Domains by Steered Molecular Dynamics Simulation. *Biophysical Journal* **75**:662-671

Manteca A, Schönfelder J, Alonso-Caballero A, Fertin MJ, Barruetabeña N, Faria BF, Herrero-Galán E, Alegre-Cebollada J, De Sancho D, Perez-Jimenez R. (2017) Mechanochemical evolution of the giant muscle protein titin as inferred from resurrected proteins. *Nat. Struct. Mol. Biol* **24**:652-657

Marko JF, Siggia ED. (1995) Stretching DNA. *Macromolecules* **28**:8759-8770

Marqusee S, Baldwin RL (1987) Helix stabilization by Glu-...Lys+ salt bridges in short peptides of de novo design. *Proc. Natl. Acad. Sci. U. S. A* **84**:8898-8902

Merkel R, Nassoy P, Leung A, Ritchie K, Evans E. (1999) Energy landscapes of receptor-ligand bonds explored with dynamic force spectroscopy. *Nature* **397**:50-53

Milles LF, Bayer EA, Nash MA, Gaub HE (2017) Mechanical Stability of a High-Affinity Toxin Anchor from the Pathogen Clostridium perfringens. *J. Phys. Chem. B* **121**:3620-3625

Milles LF, Schulten K, Gaub HE, Bernardi RC (2018) Molecular mechanism of extreme mechanostability in a pathogen adhesin. *Science* **359**:1527-1533

Minin KA, Zhmurov A, Marx KA, Purohit PK, Barsegov V. (2017) Dynamic Transition from α-Helices to β-Sheets in Polypeptide Coiled-Coil Motifs. *J. Am. Chem. Soc* **139**:16168-16177

Miserez A, Yu J, Mohammadi P. (2023) Protein-Based Biological Materials: Molecular Design and Artificial Production. *Chem. Rev* **123**:2049-2111

Muller DJ, Dumitru AC, Lo Giudice C, Gaub HE, Hinterdorfer P, Hummer G, De Yoreo JJ, Dufrene YF, Alsteens D. (2021) Atomic Force Microscopy-Based Force Spectroscopy and Multiparametric Imaging of Biomolecular and Cellular Systems. *Chem. Rev* **121**:11701-11725

Nandi T, Ainavarapu SRK (2022) Native Salt Bridges Are a Key Regulator of Ubiquitin's Mechanical Stability. *J. Phys. Chem. B* **126**:3505-3511

Neuman KC, Nagy A. (2008) Single-molecule force spectroscopy: optical tweezers, magnetic tweezers and atomic force microscopy. *Nat. Methods* **5**:491-505

Neupane K, Narayan A, Sen Mojumdar S, Adhikari G, Garen CR, Woodside MT (2024) Direct observation of prion-like propagation of protein misfolding templated by pathogenic mutants. *Nat. Chem. Biol* **20**:1220-1226

Ng SP, Billings KS, Ohashi T, Allen MD, Best RB, Randles LG, Erickson HP, Clarke J. (2007) Designing an extracellular matrix protein with enhanced mechanical stability. *Proc. Natl. Acad. Sci. U. S. A* **104**:9633-9637

Ni B, Kaplan DL, Buehler MJ (2024) ForceGen: End-to-end de novo protein generation based on nonlinear mechanical unfolding responses using a language diffusion model. *Sci. Adv* **10**:eadl4000

Ott W, Jobst MA, Bauer MS, Durner E, Milles LF, Nash MA, Gaub HE (2017) Elastin-like Polypeptide Linkers for Single-Molecule Force Spectroscopy. *Acs Nano* **11**:6346-6354

Pepelnjak M, Velten B, Näpflin N, von Rosen T, Palmiero UC, Ko JH, Maynard HD, Arosio P, Weber-Ban E, de Souza N, *et al.* (2024) In situ analysis of osmolyte mechanisms of proteome thermal stabilization. *Nat. Chem. Biol* **20**:1053-1065

Porollo A, Tien MZ, Meyer AG, Sydykova DK, Spielman SJ, Wilke CO (2013) Maximum Allowed Solvent Accessibilites of Residues in Proteins. *PLoS ONE* **8**

Randles LG, Rounsevell RWS, Clarke J. (2007) Spectrin Domains Lose Cooperativity in Forced Unfolding. *Biophys. J* **92**:571-577

Rennekamp B, Grubmüller H, Gräter F. (2024) Hidden length lets collagen buffer mechanical and chemical stress. *Phys. Rev. E* **110**:054408

Rief M, Gautel M, Oesterhelt F, Fernandez JM, Gaub HE (1997) Reversible unfolding of individual titin immunoglobulin domains by AFM. *Science* **276**:1109-1112

Rief M, Pascual J, Saraste M, Gaub HE (1999) Single molecule force spectroscopy of spectrin repeats: low unfolding forces in helix bundles. *J. Mol. Bio* **286**:553-561

Sakuma K, Kobayashi N, Sugiki T, Nagashima T, Fujiwara T, Suzuki K, Kobayashi N, Murata T, Kosugi T, Tatsumi-Koga R, *et al.* (2024) Design of complicated all-α protein structures. *Nat. Struct. Mol. Biol* **31**:275-282

Schwaiger I, Sattler C, Hostetter DR, Rief M. (2002) The myosin coiled-coil is a truly elastic protein structure. *Nature Materials* **1**:232-235

Serquera D, Lee W, Settanni G, Marszalek PE, Paci E, Itzhaki LS (2010) Mechanical Unfolding of an Ankyrin Repeat Protein. *Biophys. J* **98**:1294-1301

Shi S, Wang Z, Deng Y, Tian F, Wu Q, Zheng P. (2022) Combination of Click Chemistry and Enzymatic Ligation for Stable and Efficient Protein Immobilization for Single-Molecule Force Spectroscopy. *CCS Chem* **4**:598-604

Sivasankaran RP, Snell K, Kunkel G, Georgiou PG, Puente EG, Maynard HD (2024) Polymer-mediated protein/peptide therapeutic stabilization: Current progress and future directions. *Prog. Polym. Sci* **156**:101867

Stahl SW, Nash MA, Fried DB, Slutzki M, Barak Y, Bayer EA, Gaub HE (2012) Single-molecule dissection of the high-affinity cohesin-dockerin complex. *Proc. Natl. Acad. Sci. U. S. A* **109**:20431-20436

Takahashi H, Rico F, Chipot C, Scheuring S. (2018) alpha-Helix Unwinding as Force Buffer in Spectrins. *Acs Nano* **12**:2719-2727

Tapia-Rojo R, Mora M, Board S, Walker J, Boujemaa-Paterski R, Medalia O, Garcia-Manyes S. (2023) Enhanced statistical sampling reveals microscopic complexity in the talin mechanosensor folding energy landscape. *Nat. Phys* **19**:52-60

Thomas AS, Elcock AH (2004) Molecular Simulations Suggest Protein Salt Bridges Are Uniquely Suited to Life at High Temperatures. *J. Am. Chem. Soc* **126**:2208-2214

Tunn I, de Leon AS, Blank KG, Harrington MJ (2018) Tuning coiled coil stability with histidine-metal coordination. *Nanoscale* **10**:22725-22729

Van Der Spoel D, Lindahl E, Hess B, Groenhof G, Mark AE, Berendsen HJ (2005) GROMACS: fast, flexible, and free. *J. Comput. Chem* **26**:1701-1718

Wang Z, Nie J, Shi S, Li G, Zheng P. (2021) Transforming de novo protein a3D into a mechanically stable protein by zinc binding. *Chem. Commun* **57**:11489-11492

Watson JL, Juergens D, Bennett NR, Trippe BL, Yim J, Eisenach HE, Ahern W, Borst AJ, Ragotte RJ, Milles LF, *et al.* (2023) De novo design of protein structure and function with RFdiffusion. *Nature* **620**:1089-1100

Wensley BG, Batey S, Bone FAC, Chan ZM, Tumelty NR, Steward A, Kwa LG, Borgia A, Clarke J. (2010) Experimental evidence for a frustrated energy landscape in a three-helix-bundle protein family. *Nature* **463**:685-688

Wolny M, Batchelor M, Knight PJ, Paci E, Dougan L, Peckham M. (2014) Stable Single α-Helices Are Constant Force Springs in Proteins. *J. Bio. Chem* **289**:27825-27835

Wolny M, Batchelor M, Bartlett GJ, Baker EG, Kurzawa M, Knight PJ, Dougan L, Woolfson DN, Paci E, Peckham M. (2017) Characterization of long and stable de novo single alpha-helix domains provides novel insight into their stability. *Sci. Rep* **7**:44341

Yang R, Wong YH, Nguyen GKT, Tam JP, Lescar J, Wu B. (2017) Engineering a Catalytically Efficient Recombinant Protein Ligase. *J. Am. Chem. Soc* **139**:5351-5358

Yang S, Wang M, Tian D, Zhang X, Cui K, Lü S, H-h Wang, Long M, Nie Z. (2024) DNA-functionalized artificial mechanoreceptor for de novo force-responsive signaling. *Nat. Chem. Biol* **20**:1066-1077

Zhang X, Zhang H, Lu S, Song Y, Zhang W. (2025) Direct Detection of the Chain Trajectory and Long-Range Correlation in Crystalline–Amorphous Polymer Networks by Single-Molecule Force Spectroscopy. *Macromolecules* **58**:6170-6177

Peer reviews

Reviewer #1 (Public review):

Summary:

In the work from Qiu et al. a workflow aimed at obtaining the stabilization of a simple small protein against mechanical and chemical stressors is presented.

Strengths:

The workflow makes use of state-of-the-art AI-driven structure generation and couples it with more classical computational and experimental characterizations in order to measure its efficacy.

The work is well presented and results are thorough and convincing.

The Methods description is quite precise, and some important details were added during review.

Weaknesses:

The pulling velocity is quite high but in accordance with this observation the results were only used for comparative analyses.

Following the review process the authors have shown that the minimum distance between each protein from its periodic images was consistently above 1 nm, yet towards the end of some simulations the value crosses the non-bonded interaction cut-off distance.

Comments on revisions:

The authors did a good job in addressing the reviews.

https://doi.org/10.7554/eLife.109753.2.sa3

Reviewer #2 (Public review):

Summary:

Qiu, Jun et. al., developed and validated a computational pipeline aimed at stabilizing α-helical bundles into very stable folds. The computational pipeline is a hierarchical computational methodology tasked to generate and filter a pool of candidates, ultimately producing a manageable number of high-confidence candidates for experimental evaluation. The pipeline is split into two stages. In stage I, a large pool of candidate designs is generated by RFdiffusion and ProteinMPNN, filtered down by a series of filters (hydropathy score, foldability assessed by ESMFold and AlphaFold). The final set is chosen by running a series of steered MD simulations. This stage reached unfolding forces above 100pN. In stage II, targeted tweaks are introduced - such as salt bridges and metal ion coordination - to further enhance the stability of the α-helical bundle. The constructs undergo validation through a series of biophysical experiments. Thermal stability is assessed by CD, chemical stability by chemical denaturation, and mechanical stability by AFM.

Strengths:

A hierarchical computational approach that begins with high-throughput generation of candidates, followed by a series of filters based on specific goal-oriented constraints, is a powerful approach for a rapid exploration of the sequence space. This type of approach breaks down the multi-objective optimization into manageable chunks and has been successfully applied for protein design purposes (e.g., the design of protein binders). Here, the authors nicely demonstrate how this design strategy can be applied to successfully redesign a moderately stable α-helical bundle into an ultrastable fold. This approach is highly modular, allowing the filtering methods to be easily swapped based on the specific optimization goals or the desired level of filtering.

Weaknesses:

Assessing the change in stability relative to the WT α-helical bundle is challenging because an additional helix has been introduced, resulting in a comparison between a three-helix bundle and a four-helix bundle. Consequently, the appropriate reference point for comparison is unclear. A more direct and informative approach would have been to redesign the sequence of the original α-helical bundle of the human spectrin repeat R15, allowing for a more straightforward stability comparison.

The three constructs chosen are 60-70% identical to each other, either suggesting over-constrained optimization of the sequence, or a physical constraint inherent to designing ultrastable α-helical bundles. It would be interesting to explore whether choosing a different combination of filters would enable ultrastable α-helical bundles constructs with a more varied sequence content.

While the use of steered MD is an elegant approach to picking the top N most stable designs, its computational cost may become prohibitive as the number of designs increases or as the protein size grows, especially since it requires simulating a water box that can accommodate a fully denatured protein.

Comments on revisions:

The authors have done a good job of addressing the comments.

https://doi.org/10.7554/eLife.109753.2.sa2

Reviewer #3 (Public review):

Summary:

Qiu et al., present a hierarchical framework that combine AI and molecular dynamic simulation to design α-helical protein with enhanced thermal, chemical and mechanical stability. Strategically chemical modification by incorporating additional α-helix, site-specific salt bridges and metal coordination further enhanced the stability. The experimental validation using single-molecule force spectroscopy and CD melting measurements provide fundamental physical chemical insights into the stabilization of α-helices. Together with the group's prior work on super-stable β strands (*https://www.nature.com/articles/s41557-025-01998-3*), this research provides a comprehensive toolkit for protein stabilization. This framework has broad implications for designing stable proteins capable of functioning under extreme conditions.

Strengths:

The study represents a complete frame work for stabilizing the fundamental protein elements, α-helices. A key strength of this work is the integration of AI tools with chemical knowledge of protein stability.

The experimental validation in this study is exceptional. The single-molecule AFM analysis provided a high-resolution look at the energy landscape of these designed scaffolds. This approach allows for the direct observation of mechanical unfolding forces (exceeding 200 pN) and the precise contribution of individual chemical modifications to global stability. These measurements offer new, fundamental insights into the physicochemical principles that govern α-helix stabilization.

Weaknesses:

(1) While the initial manuscript lacked a detailed explanation for the stabilizing effect of the additional helix, the revised version now includes a clear structural basis for this improvement. The authors successfully attribute the increased unfolding force threshold to the reinforcement of the hydrophobic core and enhanced cooperative interactions, supported by relevant literature correlations between helix bundle size and stability.

(2) The author analyzed both thermal stability and mechanical stability. It would be helpful for the author to discuss the relationship between these two parameters in the context of their design. Since thermal melting probes equilibrium stability (ΔG), while mechanical stability probes the unfolding energy barriers along pulling coordinate. While the integrative

design approach successfully improved both stability types, a deeper exploration of how the specific structural modifications influence the unfolding energy barrier relative to the overall equilibrium stability would further strengthen the mechanistic impact of the work.

(3) While the current study demonstrates a dramatic increase in global stability, the analysis focuses almost exclusively on the unfolding (melting) process. However, thermodynamic stability is a function of both folding (kf) and unfolding (ku) rates. The author have clarified that the observed ultrastability likely originates from a significantly reduced unfolding rates, a hypothesis consistent with the unfolding force. Direct measurements of the kinetics would provide deeper insights.

(4) The authors chose the spectrin repeat R15 as the starting scaffold for their design. R15 is a well-established model known for its "ultra-fast" folding kinetics, with folding rates (kf ~105s), near three orders of magnitude faster than its homologues like R17 (Scott et.al., Journal of molecular biology 344.1 (2004): 195-205). Measuring the folding rates of newly designed proteins would provide additional insights into the design.

Comments on revisions:

I think the author have addressed comments.

https://doi.org/10.7554/eLife.109753.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *In the work from Qiu et al., a workflow aimed at obtaining the stabilization of a simple small protein against mechanical and chemical stressors is presented.*
>
> *Strengths:*
>
> *The workflow makes use of state-of-the-art AI-driven structure generation and couples it with more classical computational and experimental characterizations in order to measure its efficacy. The work is well presented, and the results are thorough and convincing.*

We are grateful to this reviewer for his/her thoughtful assessment and supportive feedback. In response, we have addressed each comment and incorporated the necessary revisions into the manuscript.

> *Weaknesses:*
>
> *I will comment mostly on the MD results due to my expertise.*
>
> *The Methods description is quite precise, but is missing some important details:*
>
> *(1) Version of GROMACS used.*

We used GROMACS version 2023.2 (single-precision). All subsequent MD simulation procedures mentioned below have been consolidated and described in detail in the Supporting Information (SI).

> *(2) The barostat used.*

Pressure coupling was applied using the C-rescale barostat (τ_p = 5.0 ps, ref_p = 1.0 bar).

> *(3) pH at which the system is simulated.*

No explicit pH was defined during system construction. Proteins were modeled using standard protonation states as assigned by GROMACS preprocessing tools, corresponding to physiological, near-neutral pH (~ 7.0).

> *(4) The pulling is quite fast (but maybe it is not a problem)*

The relatively high pulling velocity (1 nm/ns) was selected to enable efficient screening across a large number of designed proteins (211 candidates), while maintaining reasonable computational cost/time. Given the intrinsic orders-of-magnitude difference between simulation and experimental pulling rates, SMD results were used as a comparative screening tool, rather than for direct quantitative comparison with AFM data.

> *(5) What was the value for the harmonic restraint potential? 1000 is mentioned for the pulling potential, but it is not clear if the same value is used for the restraint, too, during pulling.*

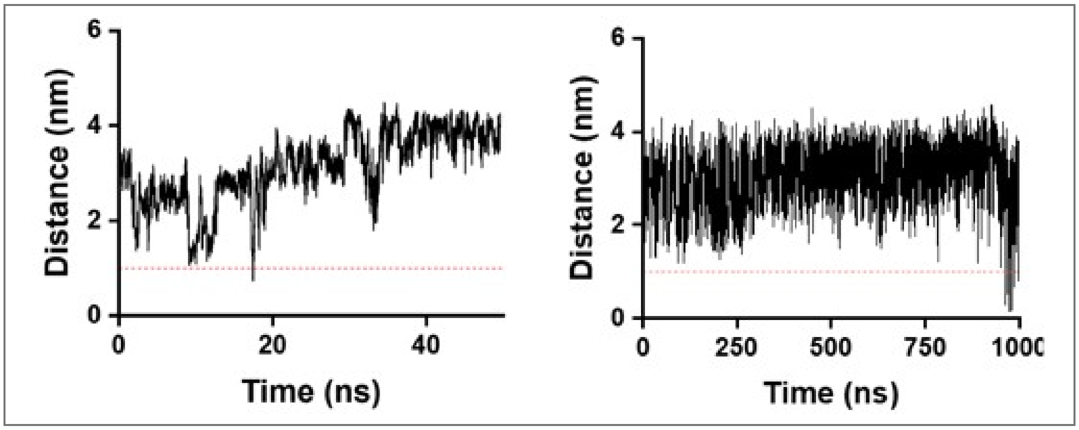
All positional restraints used in the simulations, including those applied during equilibration as well as the harmonic restraint on the N-terminus and the pulling umbrella restraint during SMD, employed the same force constant (k = 1000 kJ·mol^{-1}·nm^{2}). We have clarified this point in the revised Methods section.

> *(6) The box dimensions.*

Rectangular simulation boxes were used throughout. For equilibrium MD simulations, the box dimensions in each direction were set based on the maximum extent of the protein along that axis, with a minimum distance of 1.2 nm between the protein surface and the box boundary on all sides. For SMD simulations, the same box dimensions were applied in the x and y directions. Along the pulling (z) direction, the box length was extended to accommodate the theoretical stretching length, defined as the initial N–C terminal distance plus 0.36 nm per stretched residue, while maintaining a 1.2 nm buffer at both ends (2.4 nm total). These details have now been clarified in the revised Supporting Information.

> *From this last point, a possible criticism arises: Do the unfolded proteins really still stay far enough away from themselves to not influence the result?*

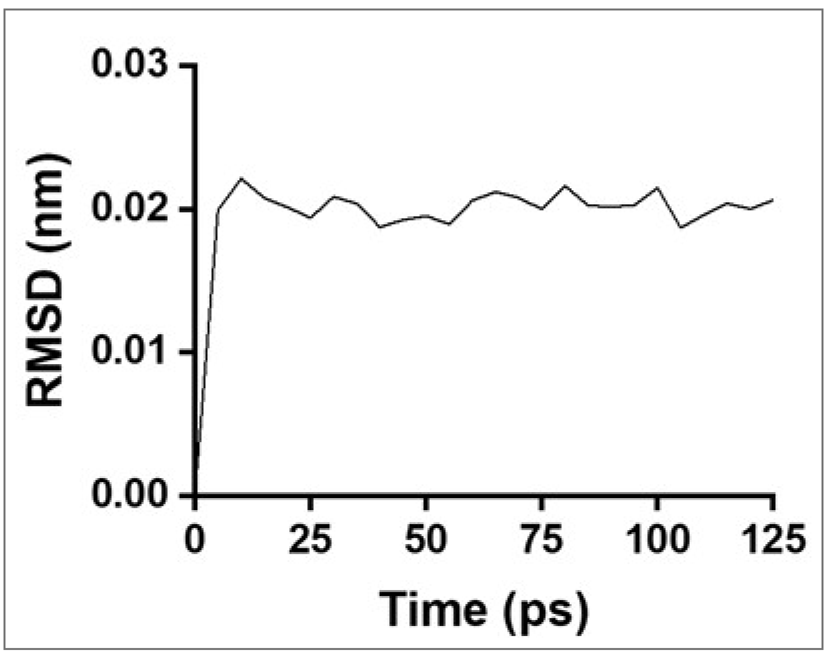
We analyzed the minimum atomic distance between each protein and its periodic images to assess potential artifacts from periodic boundary conditions. For all simulation stages used in screening and statistical analysis, the minimum protein–image separation remained above 1.0 nm for the majority of the simulation time, exceeding the nonbonded interaction cutoff and minimizing cross-boundary interactions. As shown in the Author response image 1for SpecAI89 (left), this separation during SMD simulations is consistently well above the threshold, indicating that the chosen box dimensions are appropriate. In the very late stages of annealing MD, highly unstable proteins may exhibit large conformational fluctuations and transient boundary proximity (right); however, these regimes are associated with large RMSD deviations and are excluded from analysis. Notably, the mechanically relevant unfolding events occur near the center of the simulation box and proceed along the pulling axis in SMD simulations, making boundary effects unlikely to influence the unfolding process or the relative mechanostability ranking.



Author response image 1. Analysis of the minimum atomic distance between the protein and its periodic images under periodic boundary conditions. Left: SpecAI89 during SMD simulations, showing that the minimum protein–image distance remains above 1.0 nm for the majority of the simulation time. Right: WT during AMD simulations, where transient proximity to the periodic boundary is observed at very late stages due to large conformational fluctuations.

> *Additionally, no time series are shown for the equilibration phases (e.g., RMSD evolution over time), which would empower the reader to judge the equilibration of the system before either steered MD or annealing MD is performed.*

We thank the reviewer for this suggestion. To assess equilibration, we analyzed the backbone RMSD evolution during the equilibration phase. Using SpecAI89 as a representative example (Author response image 2), the protein backbone RMSD converges rapidly and reaches a stable plateau within approximately 5 ps. The subsequent 125 ps equilibration period therefore sufficiently demonstrates that the system is well equilibrated prior to both steered MD and annealing MD simulations.



Author response image 2. The backbone RMSD of SpecAI89 over time during simulation

> ***Reviewer #1 (Recommendations for the authors):***

> *(1) In Figure S2, only one copy (or the average of the three copies; it is not clear from the caption) is shown, would be better to show the individual traces for each repeat. Additionally, only the plot for the forces is shown, and not, similarly to the AMD, the RMSD plot. This could be a stylistic choice, but it just reports on how much force was applied and not on how the protein responded to the force. Moreover, horizontal lines at the maximum value reached by the force could be added in order to directly see the difference in force applied, since it is then remarked on.*

Figure S2 originally shows a representative single SMD trajectory, as the force–extension peak positions vary between independent simulations and averaging the force traces would obscure the characteristic force peaks. In the revised Supplementary Information, we have now added the force–extension traces from the other two independent SMD repeats for each construct (New Figure S2). In addition, horizontal lines indicating the maximum force reached in each trajectory have been included to facilitate direct comparison of force differences between designs.

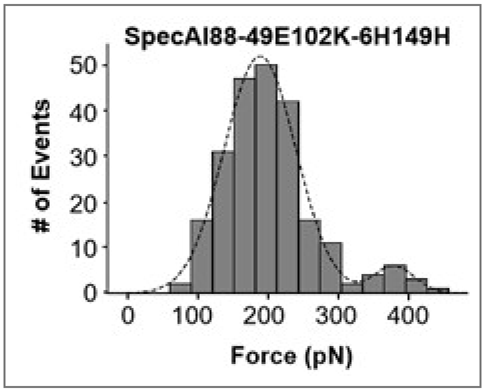
> *(2) In Figure S3 the plots have different y-axis. Maybe it could be valuable to modify it so that in figures b, c, and d the spectrum result is in the background (perhaps in gray) so that the y-axis is not changed to retain the information included in this plot, but one could still compare directly to the spectrum result. With a 0 to 1 nm y-axis part of the spectrin run will be hidden, but in any case, plot a can be used to see the full behavior. Similarly to S2, the repeats (if any) could be shown.*

We have revised Figure S3 as suggested. The y-axis is now unified to 0–1.2 nm across all panels. For panels b–d, the natural spectrin trajectory is displayed in light gray in the background for direct comparison. Additionally, three independent MD replicates are now presented for each construct to demonstrate reproducibility.

Finally, minor remarks that could nevertheless improve the paper:

> *(3) In Figure S7, a bimodal distribution model for the number of events could be used to fit the data better.*

We thank the reviewer for the detailed suggestion. Following this advice, we explored the bimodal Gaussian distribution model for fitting the force-event data in Figure S7. Indeed, our analysis showed that a bimodal fit could fit Figures S7 panel f better (as shown in Author response image 3). The two peaks were centered at $F_1 = 190 \pm 4$ pN and $F_2 = 380 \pm 6$ pN. Interestingly, the force of the first major peak obtained is the same as the previously fitted value. The second one is double force value which we guess maybe is a bi-molecule stretched for unknown reason. Considering the very few numbers of the second peak and the same force value (190 pN), we decide not to change the unfolding force value in the manuscript. But we thank this reviewer's insightful comment.



Author response image 3. **The bimodal fit for unfolding force of SpecAI88-49E102K-6H149H show the same 190 pN unfolding for the first peak as previous fit.**

> *(4) The colors in the video are not very intuitive, as the spectrin is shown initially in light blue, but becomes grey in the variants, where light blue is reserved for the additional helix. A counter of elapsed time and/or force/temperature applied could help the readers orient. Maybe it could be useful to produce a video with spectrin and the three variants all shown together?*

We thank this comment. The videos have been revised to improve clarity and consistency accordingly. In all cases, the original protein scaffold is now shown in gray, while the additional helix in the designed variants is highlighted in blue. Real-time annotations have been added to aid interpretation: the instantaneous temperature is displayed during AMD simulations, and time is shown during SMD simulations. In addition, for ease of comparison, the AMD and SMD results of all four proteins are each compiled into a single combined video, allowing their behaviors to be viewed side by side.

> ***Reviewer #2 (Public review):***
>
> *Qiu, Jun et. al., developed and validated a computational pipeline aimed at stabilizing α-helical bundles into very stable folds. The computational pipeline is a hierarchical computational methodology tasked to generate and filter a pool of candidates, ultimately producing a manageable number of high-confidence candidates for experimental evaluation. The pipeline is split into two stages. In stage I, a large pool of candidate designs is generated by RFdiffusion and ProteinMPNN, filtered down by a series of filters (hydropathy score, foldability assessed by ESMFold and AlphaFold). The final set is chosen by running a series of steered MD simulations. This stage reached unfolding forces above 100pN. In stage II, targeted tweaks are introduced - such as salt bridges and metal ion coordination - to further enhance the stability of the α-helical bundle. The constructs undergo validation through a series of biophysical experiments. Thermal stability is assessed by CD, chemical stability by chemical denaturation, and mechanical stability by AFM.*
>
> *Strengths:*

> *A hierarchical computational approach that begins with high-throughput generation of candidates, followed by a series of filters based on specific goal-oriented constraints, is a powerful approach for a rapid exploration of the sequence space. This type of approach breaks down the multi-objective optimization into manageable chunks and has been successfully applied for protein design purposes (e.g., the design of protein binders). Here, the authors nicely demonstrate how this design strategy can be applied to successfully redesign a moderately stable α-helical bundle into an ultrastable fold. This approach is highly modular, allowing the filtering methods to be easily swapped based on the specific optimization goals or the desired level of filtering.*

We are thankful for the reviewer's diligent evaluation and positive remarks. His/her concluding remarks, which encourage our future work at the intersection of AI-protein design and AFM-SMSF, are especially appreciated. All comments have been incorporated into our revisions.

> *Weaknesses:*
>
> *Assessing the change in stability relative to the WT α-helical bundle is challenging because an additional helix has been introduced, resulting in a comparison between a three-helix bundle and a four-helix bundle. Consequently, the appropriate reference point for comparison is unclear. A more direct and informative approach would have been to redesign the original α-helical bundle of the human spectrin repeat R15, allowing for a more straightforward stability comparison.*

This is an insightful comment. Indeed, a direct comparison between the same structure of the three-helix bundle will be most straightforward with a clear reference point. I will take this advice and try it in our future endeavor.

In our case, a substantial fraction of the hydrophobic region is relatively shallow and partially solvent-exposed in the wild-type R15 α-helical bundle. So, the added fourth helix provides a new hydrophobic packing interface, increasing core burial, packing density, and strengthening the internal load-bearing network. Consistent with this design rationale, rSASA analysis shows that the designed proteins exhibit a higher degree of hydrophobic core burial compared to the wild-type R15. Specifically, the fraction of residues with rSASA < 0.2 exceeds 30% in the designs, compared to 23% in the natural spectrin repeat.

> *While the authors have shown experimentally that stage II constructs have increased the mechanical stability by AFM, they did not show that these same constructs have increased the thermal and chemical stabilities. Since the effects of salt bridges on stability are highly context dependent (orientation, local environment, exposed vs buried, etc.), it is difficult to assess the magnitude of the effect that this change had on other types of stabilities.*

We agree that the effects of salt bridges are highly context-dependent and that different dimensions of stability do not always correlate. Following your suggestion, we evaluated the thermal and chemical stabilities of the Stage II constructs. The experimental results (now added as Figure S9) show that Stage II designs successfully maintain the high thermal stability and resistance to chemical denaturation to different extend. The thermal stability is still as high as the Stage I but the resistance to chemical denaturation is slightly reduced. We have added this result in the manuscript accordingly.

> *The three constructs chosen are 60-70% identical to each other, either suggesting overconstrained optimization of the sequence or a physical constraint inherent to designing ultrastable α-helical bundles. It would be interesting to explore these possible design principles further.*

Yes, the observed sequence convergence likely arises from a combination of intrinsic physical constraints of the protein architecture and the applied design and screening criteria. In particular, the tightly packed hydrophobic core imposes strong constraints on side-chain size, packing complementarity, and the alignment of heptad-like motifs reminiscent of coiled-coil organization, which collectively reduce the accessible sequence space. In addition, the strong selection pressure imposed by foldability and stability filters further promotes convergence toward similar solutions. And we agree with the reviewer that this represents an important direction for future work.

> *While the use of steered MD is an elegant approach to picking the top N most stable designs, its computational cost may become prohibitive as the number of designs increases or as the protein size grows, especially since it requires simulating a water box that can accommodate a fully denatured protein*

Yes, steered MD can become computationally expensive, particularly as the number of designs increases or as protein size grows. Considering the vast pool created by AI, SMD in this work was applied to a relatively small, high-confidence subset of candidates after multiple rounds of rapid prescreening, keeping the overall computational cost manageable. In future applications, this step could be further accelerated by integrating machine-learning–based predictors to improve scalability.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *I am not convinced that the difference in rSASA between the designs and the natural spectrin repeat is meaningful. It would be helpful to report confidence intervals for the rSASA values of the designs to clarify whether any differences are statistically robust. Even if such differences prove statistically significant, it is not clear that they are large enough to be practically meaningful.*

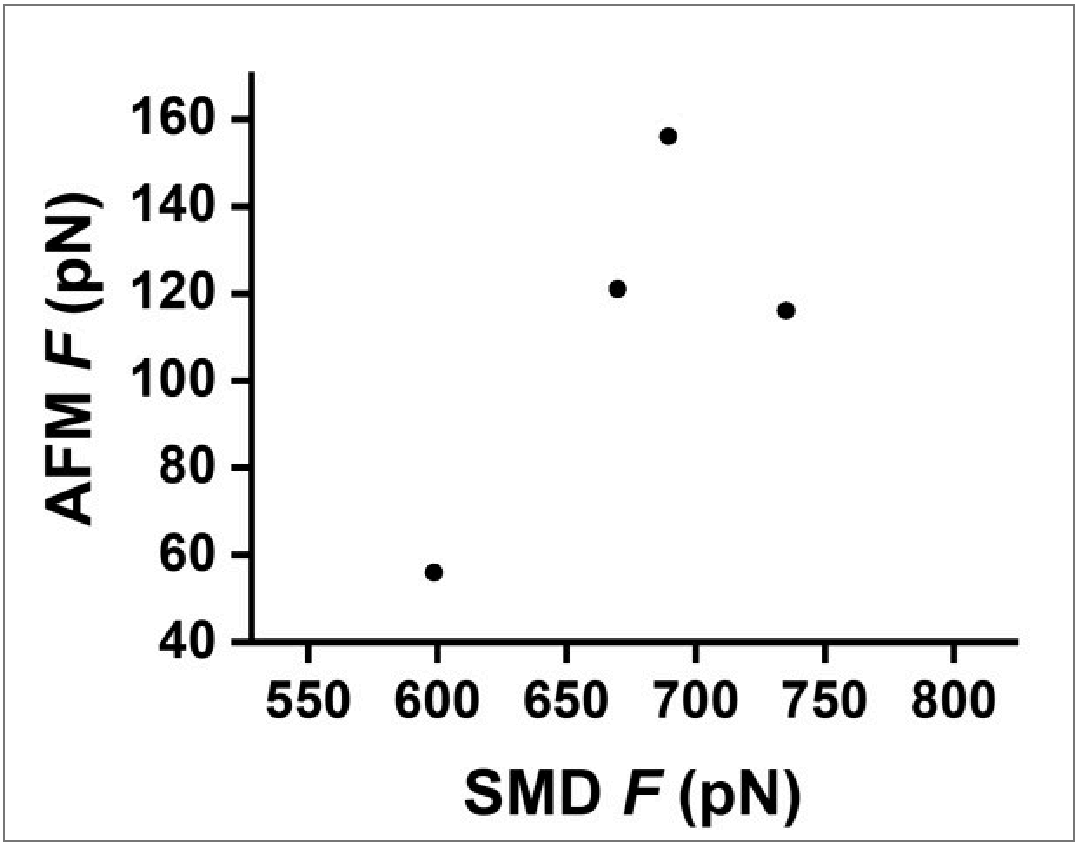
In our analysis, rSASA values were calculated from equilibrated MD conformations and were consistently higher for all designed proteins that passed the simulation-based screening compared to the wild-type spectrin repeat. However, we believe that rSASA was used only as a supportive structural descriptor to indicate a trend toward a more compact and better-buried hydrophobic core, rather than as a standalone or decisive metric of stability.

Protein stability is indeed influenced by multiple factors, including hydrogen bonding, salt bridges, metal coordination, and topology-dependent load-bearing interactions, none of which are captured by rSASA alone. Therefore, we agree with the reviewer that differences in rSASA alone should not be overinterpreted as a quantitative measure of protein stability. For this reason, rSASA was not used as a ranking criterion or a predictor of stability, but only as complementary evidence consistent with the overall design rationale and with the experimentally observed stability enhancements.

> *The claim "The strong agreement between computational rankings and experimental measurements validates this approach for prioritizing designs based on relative mechanostability, offering a practical pipeline to bridge the gap between in silico design and experimental validation." should be substantiated by a citation or a figure. Since the authors have the experimental AFM data and steered MD data, I suggest adding a Spearman correlation plot of the two.*

Following this comment, we examined the Spearman rank correlation between SMD-derived unfolding forces and experimentally measured AFM forces (Author response image 4). The resulting correlation was modest (ρ = 0.4, p = 0.6), which is not unexpected given (i) the large difference in force and timescales between high-speed SMD simulations and single-molecule AFM experiments, and (ii) the limited number of designs and simulation repeats available.

Nevertheless, qualitatively, the difference between the first point from wt-spectrin and the other three specAI is clear. Considering the large computational cost, we only performed three times simulation one each design to balance the accuracy and the cost/time. To avoid overinterpretation, we therefore did not include the correlation analysis in the main text and revised the manuscript to soften claims of strong agreement, emphasizing instead the qualitative and comparative role of SMD in the design pipeline.



Author response image 4. Spearman correlation between SMD and AFM unfolding forces for natural spectrin and SpecAI designs. SMD force (x-axis) versus experimental AFM force (y-axis); each point represents one protein.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *Qiu et al. present a hierarchical framework that combines AI and molecular dynamics simulation to design an α-helical protein with enhanced thermal, chemical, and mechanical stability. Strategically, chemical modification by incorporating additional α-helix, site-specific salt bridges, and metal coordination further enhanced the stability. The experimental validation using single-molecule force spectroscopy and CD melting measurements provides fundamental physical chemical insights into the stabilization of α-helices. Together with the group's prior work on super-stable β strands (https://www.nature.com/articles/s41557-025-01998-3), this research provides a comprehensive toolkit for protein stabilization. This framework has broad implications for designing stable proteins capable of functioning under extreme conditions.*
>
> *Strengths:*

> *The study represents a complete framework for stabilizing the fundamental protein elements, α-helices. A key strength of this work is the integration of AI tools with chemical knowledge of protein stability.*
>
> *The experimental validation in this study is exceptional. The single-molecule AFM analysis provided a high-resolution look at the energy landscape of these designed scaffolds. This approach allows for the direct observation of mechanical unfolding forces (exceeding 200 pN) and the precise contribution of individual chemical modifications to global stability. These measurements offer new, fundamental insights into the physicochemical principles that govern α-helix stabilization.*

We appreciate the positive assessment of our manuscript from this reviewer and his/her support. We have answered all the comments as follows and modified the manuscript accordingly.

> *Weaknesses:*
>
> *(1) The authors report that appending an additional helix increases the overcall stability of the α-helical protein. Could the author provide a more detailed structural explanation for this? Why does the mechanical stability increase as the number of helixes increase? Is there a reported correlation between the number of helices (or the extent of the hydrophobic core) and the stability?*

In multi-helix bundle proteins, tight interhelical packing leads to the formation of a dense hydrophobic core, which substantially enhances overall structural stability. The introduction of an additional helix does not merely increase helix count, but expands the buried hydrophobic interface, improving packing density and cooperative side-chain interactions in the core. This, in turn, strengthens the internal load-bearing network that resists force-induced unfolding.

From a mechanical perspective, adding a helix also increases topological interlocking among secondary-structure elements, which raises the energetic barrier for unfolding and shifts the unfolding pathway toward more cooperative rupture events, thereby increasing the unfolding force threshold. Consistent with this design principle, pioneering studies have reported a positive correlation between the number of helices (or the extent of the hydrophobic core) in helix bundles and their stability (Lim et al., Structure, 2008, 16:449; Minin et al., J. Am. Chem. Soc., 2017, 139, 16168; Bergues-Pupo et al., Phys. Chem. Chem. Phys., 2018, 20, 29105). Inspired by these works, our AI-protein design study uses the appended helix to reinforce the hydrophobic core rather than simply increasing secondary-structure content.

> *(2) The author analyzed both thermal stability and mechanical stability. It would be helpful for the author to discuss the relationship between these two parameters in the context of their design. Since thermal melting probes equilibrium stability (ΔG), while mechanical stability probes the unfolding energy barriers along the pulling coordinate.*

We agree this is a crucial distinction. Thermal and chemical stabilities report on the equilibrium free energy (ΔG), while mechanical stability probes the kinetic unfolding barrier (ΔG‡) along a force-dependent pathway. Their inherent difference makes concurrent improvement in all parameters a non-trivial task, which highlights the importance and success of our integrative design approach.

> *(3) While the current study demonstrates a dramatic increase in global stability, the analysis focuses almost exclusively on the unfolding (melting) process. However, thermodynamic stability is a function of both folding (k_f) and unfolding (k_u) rates. It remains unclear whether the observed ultrastability is primarily driven by a drastic*

> *decrease in the unfolding rate (k_u) or if the design also maintains or improves the folding rate (k_f)?*

We agree with the reviewer that thermodynamic stability is determined by both the folding rate (k_f) and the unfolding rate (k_u). In the present study, we did not directly measure folding kinetics, and therefore cannot quantitatively deconvolute the respective contributions of k_f and k_u to the observed ultrastability. Based on the design strategy and the experimental observations, we propose that the enhanced stability primarily originates from a substantial reduction in the unfolding rate (k_u), corresponding to an increased unfolding energy barrier. The reinforcement of the hydrophobic core, the introduction of stabilizing interactions such as salt bridges and metal coordination, and the additional helix that increases topological and packing constraints all raise the energetic cost of disrupting key interactions in the folded state.

This interpretation is consistent with the high mechanical unfolding forces observed in both AFM experiments and SMD simulations. In contrast, these stabilizing features are not necessarily expected to accelerate folding and may even modestly increase folding complexity. Addressing folding kinetics explicitly would require dedicated kinetic experiments or simulations, which are beyond the scope of the present work but represent an interesting direction for future studies.

> *(4) The authors chose the spectrin repeat R15 as the starting scaffold for their design. R15 is a well-established model known for its "ultra-fast" folding kinetics, with folding rates (k_f ~105s), near three orders of magnitude faster than its homologues like R17 (Scott et.al., Journal of molecular biology 344.1 (2004): 195-205). Does the newly designed protein, with its additional fourth helix and site-specific chemical modifications, retain the exceptionally high folding rate of the parent R15?*

We did not directly measure the folding kinetics of the newly designed proteins, and therefore cannot determine whether they retain the exceptionally fast folding rate reported for the parent spectrin repeat R15. While R15 is known for its ultrafast folding behavior, the introduction of an additional fourth helix and site-specific chemical modifications, although beneficial for enhancing stability, may increase the complexity of the folding landscape and do not necessarily guarantee that the folding rate (k_f) remains comparable to that of R15.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) Please clarify the used Gaussian function to fit the unfolding force distribution (Figure 3-4). In Figure S8, the Bell-Evans model is used to analyze unfolding force. The authors should explain the choice of fitting methods and ensure consistency.*

The Gaussian fitting used in Figures 3–4 is intended as a descriptive statistical analysis to summarize the unfolding force distributions and to facilitate direct comparison between different designs. This approach provides a robust estimate of the most probable unfolding force and the distribution width, without invoking a specific physical unfolding model, and is commonly used in single-molecule force spectroscopy for comparative purposes.

In contrast, the Bell-Evans model applied in Figure S8 is a kinetic framework that explicitly accounts for force-loading-rate dependence and is used to extract mechanistic insights into the unfolding process. Therefore, the two fitting approaches serve complementary roles: Gaussian fitting for quantitative comparison and ranking of mechanostability, and Bell-Evans analysis for mechanistic interpretation. We have clarified this distinction and the rationale for using both methods in the revised Supplementary Information to ensure consistency and transparency.

> *(2) The authors utilized steered MD simulation to analyze the mechanical properties via ForceGen (Ni et al., 2024, Sci. Adv. 10, eadl4000). However, the significant discrepancy*

> *between the predicted unfolding force (~600 pN) and the experimental value (~50 pN for spectrin, line 376) requires further justification (line 376). Please clarify how the accuracy of these predictions can be established. Specifically, do the MD simulations successfully capture the relative ranking or trends in stability across the different designed variants?*

We agree with the reviewer that there is a substantial discrepancy between the absolute unfolding forces predicted by SMD simulations (~ 600 pN) and those measured experimentally by AFM (~ 50 pN for spectrin). This difference primarily arises from the orders-of-magnitude mismatch in loading rates between simulations and experiments. In our SMD simulations, the pulling velocity (~10^9 nm/s) is several orders of magnitude higher than that used in AFM experiments (~10^3 nm/s), which is to systematically elevate the apparent unfolding force. In addition to loading-rate effects, limitations in force-field accuracy, finite system size, and restricted conformational sampling further contribute to deviations in absolute force values. As a result, the unfolding forces obtained from SMD are not intended to provide quantitative agreement with experimental measurements or absolute mechanical stability.

Instead, SMD is employed here as a comparative screening tool to assess relative mechanostability across different designed variants under identical simulation conditions. Despite the limited number of repeats imposed by computational cost, the simulations consistently distinguish candidates with markedly different mechanical responses. Importantly, the variants identified by SMD as more mechanically stable were subsequently confirmed experimentally to exhibit enhanced mechanostability relative to the wild-type spectrin repeat. Therefore, while SMD does not yield quantitatively accurate unfolding forces, it successfully captures relative stability trends and provides a practical and effective means for prioritizing designs prior to experimental validation.

https://doi.org/10.7554/eLife.109753.2.sa0